

Urban Ride-Hailing in Chicago: Descriptive, Predictive and Reinforcement Learning Approaches for Electric Fleet Operations

Advanced Analytics and Applications

Team 07



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1 Introduction

1.1 Motivation and Problem Statement

The ride-hailing market is already crowded. Uber, Lyft, FreeNow, Lime, and Tier compete for the same trips, and current market reports point to ride-hailing, increasingly paired with autonomous and electric vehicles, as the segment that will dominate urban mobility in the near term. Our client, a well-established German car manufacturer, is building a ride-hailing platform on a fully electrified vehicle fleet, wanting to launch first in the United States, where customers have proven more receptive to novel mobility business models.

The client can design and build vehicles, but has no experience running a ride-hailing fleet. Managing an electric fleet adds a further constraint: Charging schedules directly affect vehicle availability in ways a conventional fleet does not face. To plan routes, size a fleet, and decide where to deploy vehicles, the client needs a clear picture of how demand moves through a city across space and time.

This report builds from historical taxi trip data for Chicago, made available by the Chicago Data Portal and covering trips from 2024 onward. Taxis are not ride-hailing vehicles, but in a market the client has not yet entered, they are the closest available proxy for one: metered urban trips, dispatched on demand, subject to the same weather and time-of-day patterns that will govern the client's own fleet. We supplement the trip records with independently collected weather data and, where it adds value, point-of-interest data from OpenStreetMap, and we analyze demand at spatial resolutions ranging from coarse census tracts to fine-grained H3 hexagon grids, and at multiple temporal resolutions, to establish which level of detail the data can actually support. We also investigate how a vehicle should charge at a driver's home between shifts, given a battery of finite capacity and a next-shift energy requirement that is not known with certainty in advance, using Reinforcement Learning.

1.2 Research Questions

The analysis that follows is organized around five questions, each feeding into the next.

1. How does taxi demand vary across space and time in Chicago, and what plausibly explains the patterns observed at different spatial units and temporal bin sizes?
2. Where are the persistent spatial hot spots of demand, and what do Gaussian mixture models and kernel density estimation reveal about their structure?
3. Can demand be predicted for a specific spatial unit and time bucket rather than simply extrapolated from the previous hour, and how does a support vector machine compare against a feedforward neural network on this task?
4. What charging policy should an electric taxi driver follow at home, given a charging rate that can be adjusted every fifteen minutes, a battery capacity that cannot be exceeded, and a next-shift energy requirement that is only known probabilistically at the moment the driver departs?
5. What do these findings imply for the client's fleet operations, and in particular, should the client invest in private home charging infrastructure or rely on public charging networks instead?

2 Data Description

2.1 Data Sources

Three raw datasets underpin the analysis: taxi trip records, hourly weather observations, points of interest, accompanied by a boundary layer of Chicago’s community areas. Each is described below by origin, coverage, and native granularity, the cleaning and feature-engineering steps applied to each follow in Section 3.

The taxi trip records come from the Chicago Data Portal’s Taxi Trips dataset, which publishes one row per dispatched, metered taxi ride in the city. The raw download spans January 1, 2024 through May 31, 2026 and holds 16,086,255 trips across 23 fields: trip and taxi identifiers, start and end timestamps, duration and distance, the fare broken into its components and as a total, the operating company, and pickup and dropoff locations given as census tract, community area, and centroid coordinates. Locations are masked for privacy, so a published coordinate is the centroid of the containing census tract rather than the true pickup point, and the tract itself is withheld on just over half of all trips (55.3% of pickups, 56.6% of dropoffs) against 2.8% and 8.8% for community area. Each trip is a ride requested wherever and whenever a passenger wanted one, which is the demand pattern the client’s own fleet will have to anticipate.

Weather observations come from the ERA5-Land reanalysis product, pulled through the Copernicus Climate Data Store API for a single coordinate near Chicago (latitude 42, longitude -87.8) rather than a network of ground stations. The record runs from January 1, 2024 through May 12, 2026 at hourly resolution and carries six variables: 2-metre air temperature, total precipitation, snow cover, snow depth, and the two horizontal components of wind at 10 metres. A reanalysis combines numerical weather prediction models with observational data through data assimilation, resulting in a temporally continuous representation of atmospheric conditions, which no single Chicago weather station could promise.

Points of interest come from OpenStreetMap, extracted with OSMnx for the “Chicago, Illinois, USA” boundary and limited to three tag families: amenity, shop, and tourism. The raw download returns 43,237 points. Unlike the trip and weather data, this is a snapshot of the map as it stands today, not a historical record tied to the trip period. The boundary layer for Chicago’s 77 official community areas is used for spatial reference and choropleth mapping throughout the report. It is a distinct unit from the census tracts already present in the trip data itself.

3 Data Preparation

This section describes the preparation pipeline that turns the raw sources described in Section 2 into the analysis-ready datasets used throughout the report. A detailed table of the engineered features can be seen in the Table 3 in the appendix.

3.1 Taxi Trip Description and Cleaning

Cleaning the raw trip table (Section 2) proceeds in three stages: trimming columns, removing unusable rows, and splitting the result into the two parallel datasets the rest of the pipeline consumes.

Seven of the 23 raw fields are dropped as redundant. The individual fare components (fare, tips, tolls, and extras) are already summed into the trip total, the payment type says nothing about where or when demand appears, and the two point-geometry columns merely repeat the latitude and longitude stored separately. Sixteen fields remain: trip and taxi identifiers, start

and end timestamps, duration and distance, pickup and dropoff census tract and community area codes, the fare total, the operating company, and pickup and dropoff coordinates. The export writes decimals with commas and prefixes currency with a dollar sign, so these columns are parsed locale-aware before any numeric comparison is made on them. From the coordinates we also derive an H3 cell index per trip at resolutions 7, 8, and 9, which gives the hexagonal units used alongside tracts and community areas later.

Rows are removed on two grounds. Records missing a taxi identifier, an end timestamp, a duration, a distance, or a fare total are dropped outright. Each of these is absent on well under 0.2% of trips, a share small enough that imputation would add assumptions without adding information. A sanity filter then removes trips with a non-positive duration or distance, or a negative total, none of which describe a journey that actually happened. No statistical outlier removal is applied beyond that: extreme but plausible values, such as long airport transfers, are legitimate demand and are deliberately kept.

Because census tract is withheld far more often than community area, the cleaned data is kept as two parallel datasets rather than forcing one compromise on every downstream analysis. The larger retains every trip with a known community area, the stricter keeps only those that also carry a census tract. Both are truncated to the last hour the weather series covers, so that no model can later be trained or evaluated against forward-filled weather. That leaves **12,949,755 trips at community-area completeness** and **6,052,512 at census-tract completeness**, both spanning January 1, 2024 through May 12, 2026.

3.2 Weather Data

The final dataset comprises 17,544 hourly observations across nine columns (a timestamp, six raw variables (Table 1), and two engineered wind variables) with no missing values, consistent with the reanalysis nature of the data. A small number of precipitation and snow cover values were marginally negative due to rounding errors in the underlying computations and were clipped to zero. Temperature and precipitation were additionally converted to Celsius and millimeters for interpretability. Exploratory analysis confirms clear seasonal patterns, with snow cover and snow depth peaking between November and March/April and falling to near zero in summer. Snow cover approaches 100% precisely when snow depth peaks and temperatures are lowest, confirming internal consistency, while precipitation shows no strong correlation with snow variables, as expected given their distinct physical origins. No anomalies or implausible extremes were found. The temperature ranges from below -30°C to roughly 35°C , precipitation stays below ~ 20 mm/hour, and snow depth does not exceed 30 cm which is consistent with Chicago’s historical climate. For the client, this means the weather inputs for our models are trustworthy and reliable to use.

3.3 Points of Interest

Point-of-interest data was optional under the assignment brief, but there is precedent for treating it as a demand signal: Yan et al. (2020) predict taxi demand at city scale by combining several sources of urban geospatial data, point-of-interest records among them. We therefore include POI density to test how much it explains about where and when ride demand originates in Chicago. Coverage across the three tag families is uneven: 7,053 records lack an amenity tag, 37,552 lack a shop tag, and 41,738 lack a tourism tag, so most points carry only one of the three. We collapsed them into a single `poi_type` field by taking amenity where present and falling back to shop, then tourism, which left every point with one of 311 distinct raw types and none without a type or a geometry.

At 311 types the data is too granular to serve directly as model features, so we mapped each raw type onto one of fourteen broader categories that plausibly drive ride-hailing demand: food and drink, nightlife, entertainment, leisure and sports, grocery, shopping, health, finance, education, lodging, transport, automotive, services, and civic and community. The mapping doubles as a filter. Fifty-five raw types like infrastructure and street furniture (benches, parking entrances, post boxes, waste bins), fit no category and were dropped, leaving 20,108 categorised points of interest. Roughly half of those are building polygons rather than single points, so each was reduced to its centroid and assigned to a spatial unit, then aggregated to all three units used downstream: H3 resolution 7, census tract, and community area. The handful of points that fall just outside the official city boundary are excluded from the tract and community aggregations but retained at the H3 level, which is defined continuously over any coordinate.

3.4 Spatial and Temporal Discretisation

The grid is built from H3 resolution 7, every hexagon whose centroid falls inside the Chicago boundary. Census tract and community area join the same lookup by matching each hexagon centroid against the tract and community polygons, giving one crosswalk across all three spatial units. Every unit is then crossed with two temporal bins, hourly and daily, with pickups aggregated against all spatial and temporal levels. Spatial units that never register a single pickup over the full 2.4-year window are dropped before the grid is written. The reduction removes only all-zero rows and costs no trips.

4 Data Analytics

This section covers all analytical methods applied to the prepared data: descriptive spatial analytics, predictive modelling of demand (SVM and neural networks), and reinforcement learning for smart charging.

4.1 Descriptive (Spatial) Analytics

To better understand the characteristics of taxi demand before developing predictive models, we first perform a descriptive analysis of the dataset. The analysis investigates spatial and temporal demand patterns, compares different spatial representations, and identifies persistent demand hot spots.

4.1.1 Spatial Patterns

Taxi demand was analysed using community areas, census tracts and H3 hexagonal grids at multiple spatial resolutions. While all representations reveal the same overall demand structure, the choice of spatial unit strongly affects the level of detail. Community areas provide a coarse overview, whereas census tracts capture much finer local variation. H3 resolution 7 offers a good compromise between spatial detail and data density, while H3 resolution 8 provides little additional information because the published coordinates are masked to census tract centroids.

Demand is highly concentrated across all spatial representations. The Lorenz curves (Figure 10) show that only a small fraction of spatial units accounts for the majority of all taxi trips, indicating a strong spatial imbalance. Increasing the spatial resolution reveals more localized demand patterns but also substantially increases sparsity. For example, 96.1% of census tract hour-cells and 89.2% of H3-9 hour-cells contain no trips, whereas community areas remain considerably denser (45.2% empty cells). H3 resolution 7 therefore provides a good trade-off between spatial detail and data availability and is chosen as the H3-Resolution used for the

predictive analysis. Spatial demand maps, Lorenz curves, sparsity statistics and additional spatial analyses are provided in Appendix A (Figure 6–Figure 10, Figure 9 and Table 2).

4.1.2 Temporal Patterns

Taxi demand exhibits pronounced daily and weekly patterns. Demand is lowest during the early morning hours, increases rapidly from around 6 a.m., and remains high throughout the afternoon before declining again during the night. Weekday demand is characterized by pronounced commuting peaks, whereas weekend demand shifts towards the late evening and nighttime hours.

The daily demand series further reveals a stable weekly seasonality throughout the observation period, while demand levels gradually increase from early 2024 into 2025 before remaining relatively stable. Short-term drops are visible around holiday periods, most notably at the end of each year.

Trip characteristics also vary systematically over the day. Average trip distances are largest during nighttime when traffic volumes are lower, whereas average travel speeds decrease during periods of high demand, reflecting congestion during commuting hours. Approximate operational idle time follows the opposite behaviour, reaching its minimum during the morning demand peak and increasing during periods of lower demand. Additional temporal analyses are shown in Appendix A (Figure 13–Figure 20).

4.1.3 Demand Hot Spots

Persistent demand hot spots were identified using Gaussian Mixture Models (GMM) and Kernel Density Estimation (KDE). Both methods consistently identify the Chicago central business district as the dominant concentration of taxi demand, while additional demand centres emerge around O’Hare Airport and Midway Airport. The KDE highlights a continuous demand corridor extending through the city centre, whereas the GMM partitions the city into regions with similar demand characteristics.

A comparison with the aggregated POI data shows that the identified clusters correspond closely to different urban functions. Airport clusters are dominated by transportation-related POIs, while central clusters contain high densities of restaurants, retail and commercial establishments. Residential areas exhibit substantially lower demand densities. These findings indicate that land use is a strong determinant of taxi demand and motivate the inclusion of spatial context features in the predictive models. The hotspot visualizations are provided in Appendix A (Figure 11 and Figure 12).

4.2 Predictive Analytics (SVM & Neural Networks)

Fleet planning needs to know where and when demand will appear before a shift starts. We predict trip starts per spatial unit and time bucket from the cell’s identity and location, calendar, weather, and POI features alone, a scenario question such as expected demand on a rainy Sunday morning on the outskirts, not an extrapolation from the previous hour. Two model families are compared: a Support Vector Machine, fitted directly on demand and on the residual over a historical-mean baseline, and a feedforward neural network. Both are trained across six grids combining three spatial units (H3 resolution 7, census tract, community area) with hourly and daily bins, and evaluated against the same historical-mean baseline to assess whether a learned model adds value over simply replaying past averages.

The target is **Total_Trip_Start**, the trip count per spatial unit and time bucket. Train, validation, and test are cut along the calendar rather than drawn at random: the earliest 50% of unique dates train the models (2024-01-01 to 2025-03-06), the next 20% validate (2025-03-07 to 2025-08-26), and the final 30% are held out for testing (2025-08-27 to 2026-05-12). A random row-level split would leak autocorrelated demand into the test set and contaminate **base_demand**. Model selection runs on validation and test is scored once, for the selected models. The reference point throughout is a historical mean per spatial unit, grouped by month and day of week, with hour added on the hourly grids. This baseline is computed on the training split alone and used in two ways: as the **base_demand** feature passed to both models, and as the denominator in the skill score $1 - \text{MAE_model} / \text{MAE_baseline}$. We report skill rather than raw MAE because mean demand differs by a factor of roughly 80 across the six grids, making absolute errors hard to compare between them.

4.2.1 Support Vector Machine

Two Support Vector Machines are trained on the same data, differing only in their prediction target. The direct model predicts trip demand directly. The residual model predicts how far a given place and time departs from its own historical average, with that correction then added back to the average. The motivation is that the historical average already captures most of the predictable signal, so a model trained to correct it can focus on the remaining variation rather than relearning what the average already provides.

Both models share the same training setup. Because most spatial and temporal cells contain no trips, each model trains on a balanced mix of zero and non-zero rows so it does not simply learn to predict zero, while validation and test use the natural, unbalanced proportions. For each model we compare four feature sets, starting from a calendar and location baseline and extending to a version that adds weather and points of interest, and search for the best model settings. The exact search grid and feature blocks are listed in Table 4 and Table 5.

4.2.1.1 Validation Results

The direct model does not beat the historical-mean baseline at most levels. Its tuned winner trails the baseline on validation at five of the six grids, and the one exception produces only a marginal gain. This follows from the prediction target: predicting absolute demand requires the model to reproduce the historical pattern before it can improve on it, and most of the six grids do not provide enough signal for both (Table 6, Figure 21).

Predicting the correction to the historical mean rather than demand itself improves results at all six levels, though not evenly. Community/daily shows the largest gain, H3_7/daily a smaller but consistent one. Three levels land close to the baseline in either direction, and census/hourly is the only level where the residual model performs worse than the direct one. In general, the SVM performs better on daily data than on hourly data, under both strategies (Table 7, Figure 23, Table 8).

The two strongest configurations overall are both residual models fitted on daily data, and both do well with a simple linear fit rather than a more complex one. The margin they add over the historical average is real but modest, with the exact figures in Table 7.

4.2.2 Deep Learning Model

The neural network differs from the SVM in two ways. It trains on the full, unbalanced training data, as a balanced sample produced worse results in initial tests. It also uses a single

feature set at every level, combining all available features, as comparing four sets was not feasible within the available compute budget. Architecture and training settings are selected by a grid search per level using validation MAE, with the full search grid in Table 9, and the winning configurations in Table 10.

4.2.2.1 Validation Results

The network performs well at hourly resolution but not at daily. It beats the historical-mean baseline at only two of the six levels, both on hourly data. Its best result is a 9.7% skill gain at community/hourly, which is also the level where the SVM performs worst. Every daily-resolution grid falls below the baseline regardless of spatial unit, with h3_7/daily the weakest at nearly 32% below it. At daily resolution, the substantially smaller number of training observations makes it harder for the network to learn meaningful patterns from the data. Accordingly, temporal resolution matters more than spatial granularity for this model (Table 11, Figure 25).

The selected architectures vary by level without following a simple pattern. Community/hourly uses the largest and deepest network tested. The two census levels, the sparsest in the dataset, both select the smallest. Dropout is selected only at h3_7/daily and census/daily, the two levels most prone to overfitting (Table 10). This leads to the following conclusion: The historical baseline already captures most of the predictable signal, and daily grids leave little room for the neural network to improve on.

4.2.3 Model Comparison

The two model families are compared on the same held-out test data, at the level where each independently performs best: Chicago’s 77 community areas, assessed separately at daily and hourly resolution. Both models perform best at this comparatively coarse geography, with neither the hexagon grid nor individual census tracts producing more accurate forecasts for either model. A citywide rollout does not need finer-grained location data to plan around.

Neither model outperforms the other across all settings. At daily resolution, the SVM reduces the average error from 49.58 to 46.34 trips per area per day, a 6.5% improvement over the historical-mean baseline, while the neural network performs 10.0% worse than that same baseline. At hourly resolution the result is reversed: the neural network reduces the average error from 2.90 to 2.62 trips per area per hour, a 9.7% gain, while the SVM performs no better than the baseline (Table 12). For the client, this means the SVM is the better choice for day-ahead capacity planning, and the neural network for hourly demand forecasting. Therefore, both models should be used accordingly.

Breaking down results by demand level adds further detail (Table 13). At daily resolution, the SVM beats the baseline across every demand tier, with its largest gain of 7.0% on high-demand days. The neural network falls below the baseline at every daily tier, with the largest gap on medium-demand days at more than 30%. At hourly resolution, the SVM performs well only on zero-demand hours, with a 19.4% gain, but performs below the baseline on low-demand and high-demand hours. The neural network beats the baseline on every hourly tier with any demand, gaining 11.3% on medium-demand hours and 11.1% on high-demand hours. As shown in (Figure 34), both models follow a similar pattern of relative error across the day, with higher values in the early morning hours (0–4), where mean demand is lowest. As shown in (Figure 31), relative error increases towards the weekend for both models and both temporal resolutions.

A forecast that only works in good weather would be of limited use for operational planning. Neither model shows this weakness. The SVM’s daily accuracy is slightly higher on rainy days

than on dry ones, and the neural network’s hourly accuracy follows the same pattern. Neither model degrades noticeably in rain at this level of geography.

The spatial distribution of errors is consistent with the demand-tier results (Figure 32). At daily resolution, the SVM produces lower errors across almost every community area. At hourly resolution, the neural network’s advantage is concentrated in the busiest central districts, downtown and the Near North Side, where hour-to-hour demand varies the most. Further breakdowns by day of week, hour of day, and calendar period are collected in Section A. Both models track the underlying seasonal demand pattern across the full test period at both temporal resolutions, as shown in (Figure 35).

4.2.3.1 Recommendation

The two models are best deployed together: the SVM for day-ahead capacity planning across service areas, and the neural network for hourly demand forecasting within a shift. Since a finer geographic grid did not improve accuracy for either model, investing in more granular location data is not a priority at this stage.

4.2.4 Shortfalls and Improvement Levers

Both models share the same ceiling. The historical average already captures most of the predictable signal for a given place and time, and neither model improves on it by a wide margin. On the sparsest grids, too few trips occur for either model to add meaningful value beyond the baseline. For the SVM, training on the full, unbalanced data rather than the balanced sample used, would likely address its remaining weakness on high-demand hours. For the neural network, the grid search was limited by available compute resources. Further exploration of layer configurations, AdamW parameters, and alternative activation functions could improve results.

4.3 Smart Charging (Reinforcement Learning)

The final task moves from forecasting demand to acting on it: an automated controller that decides how fast an electric taxi should charge at home between shifts, balancing the electricity bill against the risk of leaving too little charge for the next day’s work. We compared two reinforcement-learning methods, Tabular Q-Learning and a Deep Q-Network (DQN). The simpler Tabular approach is the one we recommend deploying now. DQN is held back for later, more complex versions of the problem, for the reasons that follow.

4.3.1 Problem Framing & Methodology

Each vehicle returns home at 2:00 PM and must depart at 4:00 PM. During this window, an automated controller decides every 15 minutes how fast to charge the battery, choosing from four charging speeds. Electricity prices rise later in the afternoon, while the energy required for the day’s shift is unknown until departure and varies around 30 kWh. Running out of energy mid-shift causes a missed fare and a stranded vehicle is substantially more costly than a marginally higher electricity bill. Thus, the controller must balance cost against this risk. The problem was formalized as a Markov Decision Process (MDP). At each step, the model observes battery level and remaining time, and selects one of four charging speeds (0, 7.33, 14.67, or 22 kW). Each decision incurs a small electricity cost, and a \$500 penalty applies if the battery is insufficient at departure. This penalty was set far above the maximum achievable savings from smart charging (at most a few cents per day) so that the model always prioritizes reliability. The arrival battery level is drawn from a uniform distribution (0–50% of capacity) rather than

assumed as always empty, reflecting realistic driver behavior and creating a genuine cost-risk trade-off. The per-kWh price coefficient rises from ~ 7 to ~ 11 cents across the eight time steps, incentivizing earlier charging where safe. Two methods were compared: Tabular Q-Learning, a lookup table well suited to this problem’s small, enumerable state space, and DQN, which replaces the table with a neural network capable of generalizing across states. Each method was trained independently five times with different random seeds and evaluated on several thousand simulated days per run, to avoid drawing conclusions from a single, potentially lucky training outcome.

4.3.2 Results

4.3.2.1 Tabular Q-Learning

Training over 150,000 simulated days shows the training score improving and then stabilizing, with the failure rate dropping sharply to a low, stable level (Figure 37, Figure 38), indicating convergence to a consistent policy. The learned policy (Figure 39) uses both intermediate charging speeds across a substantial share of states rather than only the extremes, indicating a genuinely differentiated strategy. Tabular Q-Learning reduced the average daily cost from \$1.92 to $\$1.50 \pm \0.02 while keeping the failure probability ($0.11\% \pm 0.09\%$) close to the naive baseline (Table 14). Overall, a roughly 22% cost reduction without a measurable increase in stranded-vehicle risk.

4.3.2.2 Deep Q-Network

DQN’s training curves (Figure 40, Figure 41) oscillate throughout the full 20,000-episode period rather than stabilizing, so the best-performing weights observed at any point were retained rather than the final ones. Increasing episodes further did not help, which is also why DQN required far fewer episodes than the tabular approach (20,000 vs. 150,000), reducing the compute time needed drastically. The retained policy (Figure 42) charges at full power in almost all states, differing from the naive baseline in only two narrow state regions. Largely reproducing the naive strategy rather than a materially different one. The DQN heatmap appears fully populated while the Q-table heatmap only covers a diagonal band, because the Q-table stores entries only for visited states, whereas the neural network returns an output for any input, including states never encountered during training. DQN reduced the average daily cost to $\$1.46 \pm \0.24 with a failure probability of $1.16\% \pm 0.96\%$ which is comparable on average to Tabular Q-Learning, but with much greater variability across runs (Table 15). The per-seed variation is one of the key findings: one run reproduced the naive strategy almost exactly, while the other four achieved lower costs at markedly higher failure rates (Table 16). This inconsistency, more than the average result, is why DQN is not recommended for production use at this stage, as we are able to achieve comparable or even better results with noticeably less effort.

5 Discussion and Outlook

Taxi demand across Chicago is concentrated, not spread. At every spatial resolution tested, roughly 99% of pickups occur in the busiest tenth of locations, and hot-spot analysis places that concentration where the densest mix of commerce, transit, and evening entertainment sits: the central business district and the major transport hubs. Demand also follows a dependable daily rhythm, quiet overnight and peaking around the commute and again in the evening, with weekends distinct from weekdays. This is structure a fleet operator can plan against. Moving to a finer geographic grid added little beyond what census tracts and community areas already showed, because the data’s privacy masking snaps most coordinates back to their census tract.

For the client, that points to concentrating the initial fleet in the few districts that generate almost all demand rather than spreading vehicles evenly, and to planning around the administrative geography Chicago already publishes rather than a bespoke fine-grained grid.

Demand can be predicted for a specific place and time, not just extrapolated from the previous hour, though within limits. Both models are most accurate at the same geography, Chicago's 77 community areas, and neither wins outright. They divide the work by time horizon. The support vector machine is the more reliable day-ahead forecaster, beating a historical-average benchmark by about 6.5% and holding that edge on quiet and busy days alike. At hourly resolution, the neural network outperforms the benchmark by around 9.7%, with the gap being most visible during peak hours where the SVM performs less well. Notably, neither model struggles on rainy days. At daily resolution, however, the neural network performs worse than even the naive benchmark. Running both models in combination therefore seems more practical than choosing one: the SVM for day-ahead planning and the neural network for adjustments within a shift.

For smart charging, Tabular Q-Learning is the method to deploy first. It converged reliably across every training run and cut the average daily charging cost by roughly 22% against the naive baseline, while holding the risk of running out of charge near that baseline and so keeping vehicle availability intact. The Deep Q-Network matched it on average cost and failure rate but varied far more from run to run, a poor trade for a service where one stranded vehicle means a missed fare. It becomes worth revisiting only as the fleet and its charging decisions grow more complex.

That same controllability settles the larger infrastructure question. Both the cost saving and the reliability depend on dictating when and how fast each vehicle charges, which is only possible on chargers the operator controls. Private charging at the driver's home, over the idle window between shifts, is therefore the better default than leaning on public stations, where the driver pays the posted rate and competes for an open plug. Public charging is best reserved as a fallback, for unusually long shifts or drivers without home access. This rests on the operational case rather than a full comparison of installation cost against public-charging fees, which a follow-up should price out.

The results should be interpreted with some caution. They are based on Chicago taxi trips rather than the client's own electric fleet, and the demand models only modestly outperform a simple historical average, which suggests that location and calendar already account for most of the predictable variation. The charging result is also limited in scope: it assumes a fixed daily price curve and a single vehicle, not a full fleet in a real market. Once the client begins operating, validating these results against actual fleet data would be an important next step.

A Additional Figures and Tables

This appendix collects supporting figures and tables that are not reproduced in the main body of the report, organized by the notebook that produces them (see the pipeline overview in the project `CLAUDE.md`).

A.1 Weather Data Preparation (1.2)

Table 1: Selected Meteorological Variables

Variable	Description	Datatype
2m Temperature (K and °C)	Air temperature at 2 meters above the surface. Expected to drive customer demand (e.g., fewer walk/bike substitutes in extreme cold) and EV battery performance, since range may drop at low temperatures.	float64
Total Precipitation (m and mm)	Accumulated liquid and solid precipitation reaching the surface. Expected to increase short-distance ride demand and reduce walking/cycling alternatives.	float64
10m Wind Components u10 and v10	Eastward and northward horizontal wind components at 10 meters above ground level. Combined into wind speed and direction (see Feature Engineering); relevant for extreme-weather demand spikes and outdoor-wait discomfort.	float64
Snow Cover / Snow Depth	Fraction of the grid cell covered by snow, and mean snow thickness on the ground (excluding canopy snow). Expected to directly affect road conditions, trip duration, and safety-related demand drops, and are key drivers of charging infrastructure planning in winter months.	float64

A.2 Points of Interest Data Preparation (1.3)

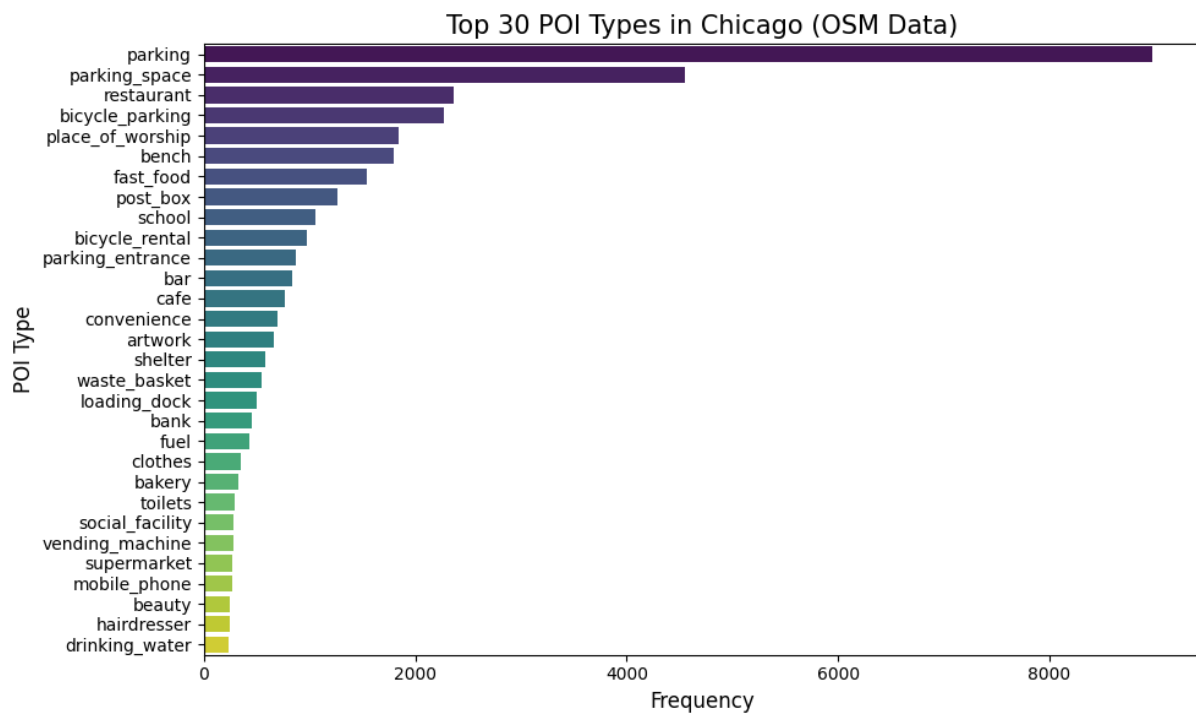


Figure 1: Top 30 most frequent raw OSM POI types in Chicago, before category mapping.

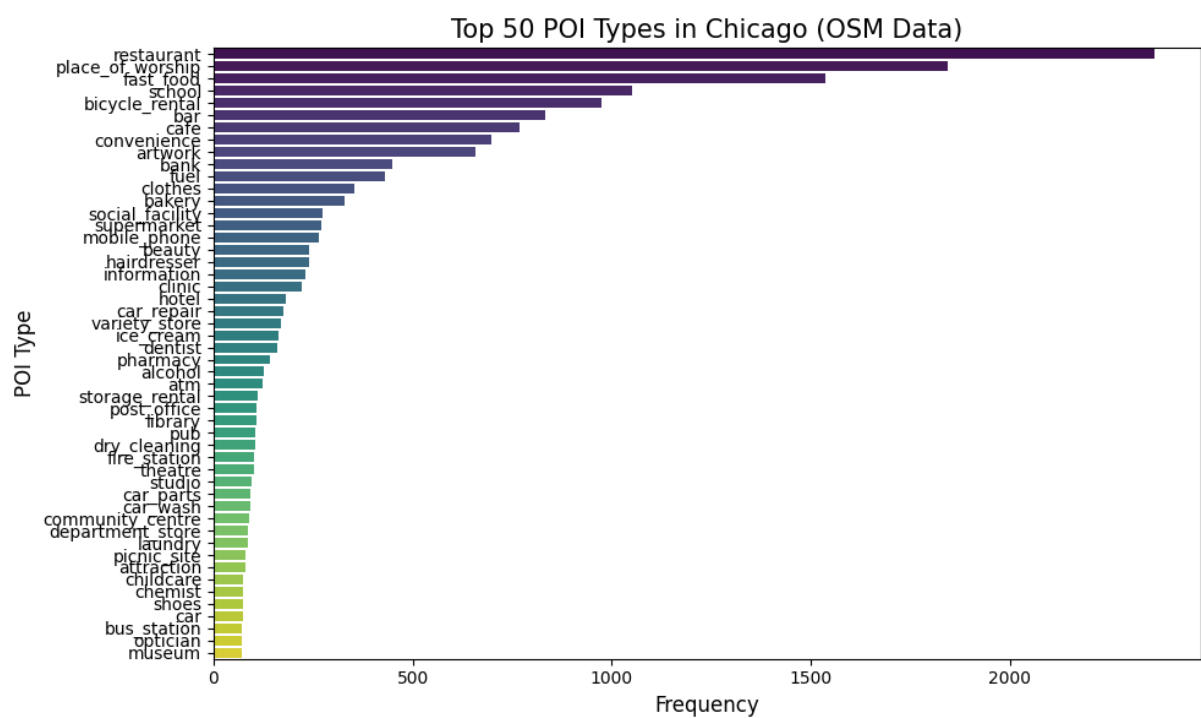


Figure 2: Top 50 most frequent POI types remaining after category filtering (unmapped ‘other’ types dropped).

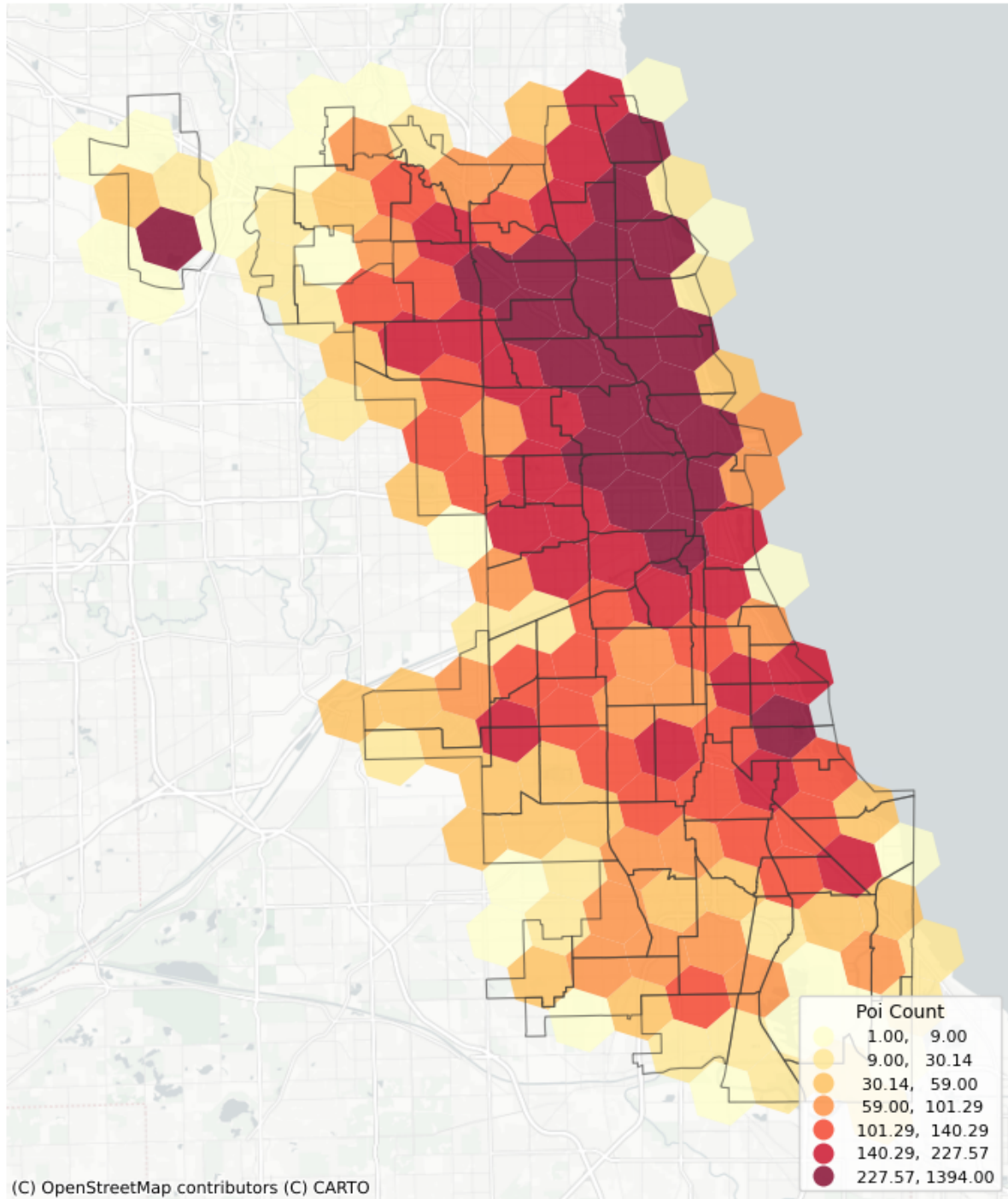
Chicago — POI Count by H3 Hexagon (resolution 7)

Figure 3: Total POI count per H3 hexagon (resolution 7).

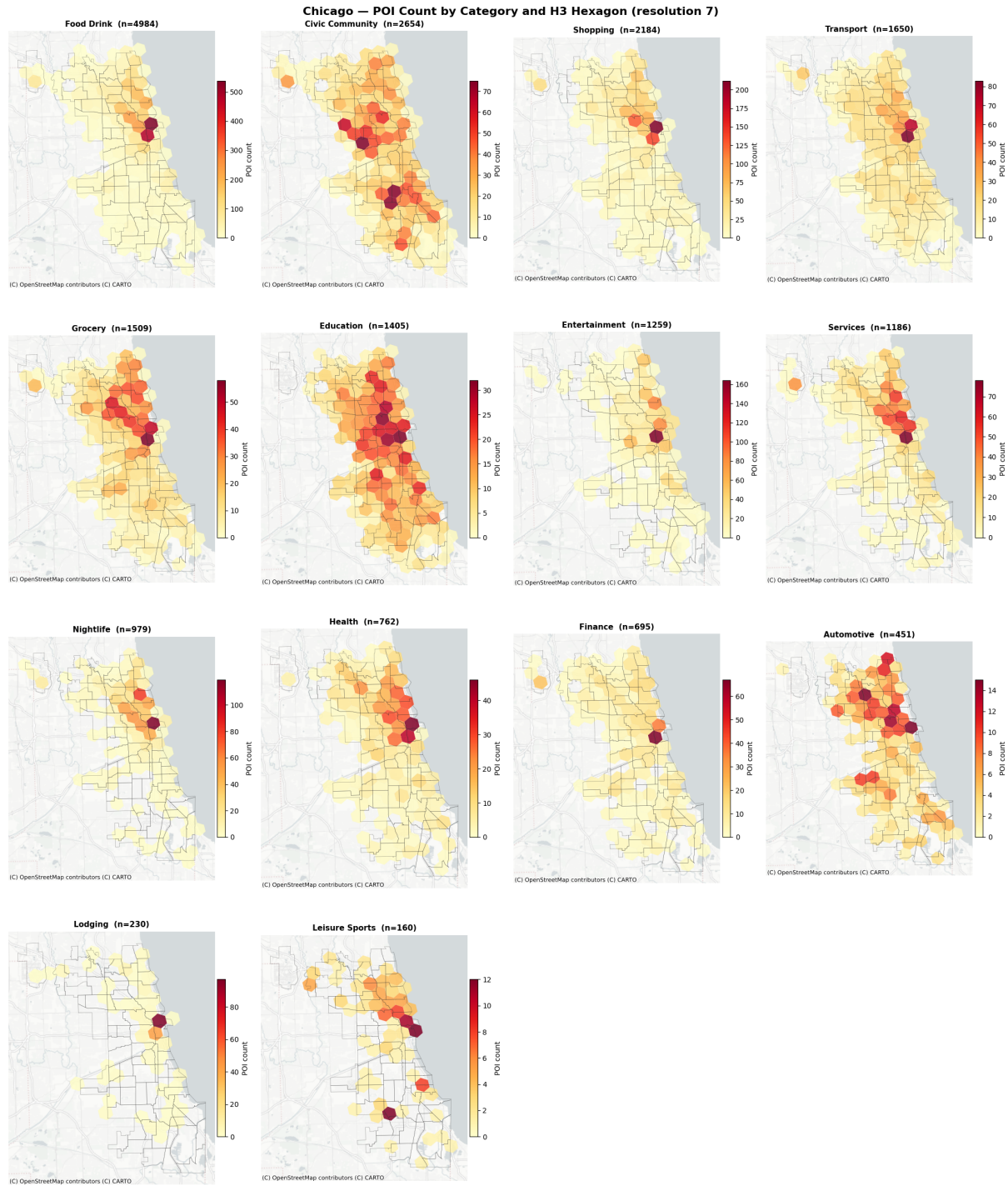


Figure 4: POI count per H3 hexagon (resolution 7), faceted by category.

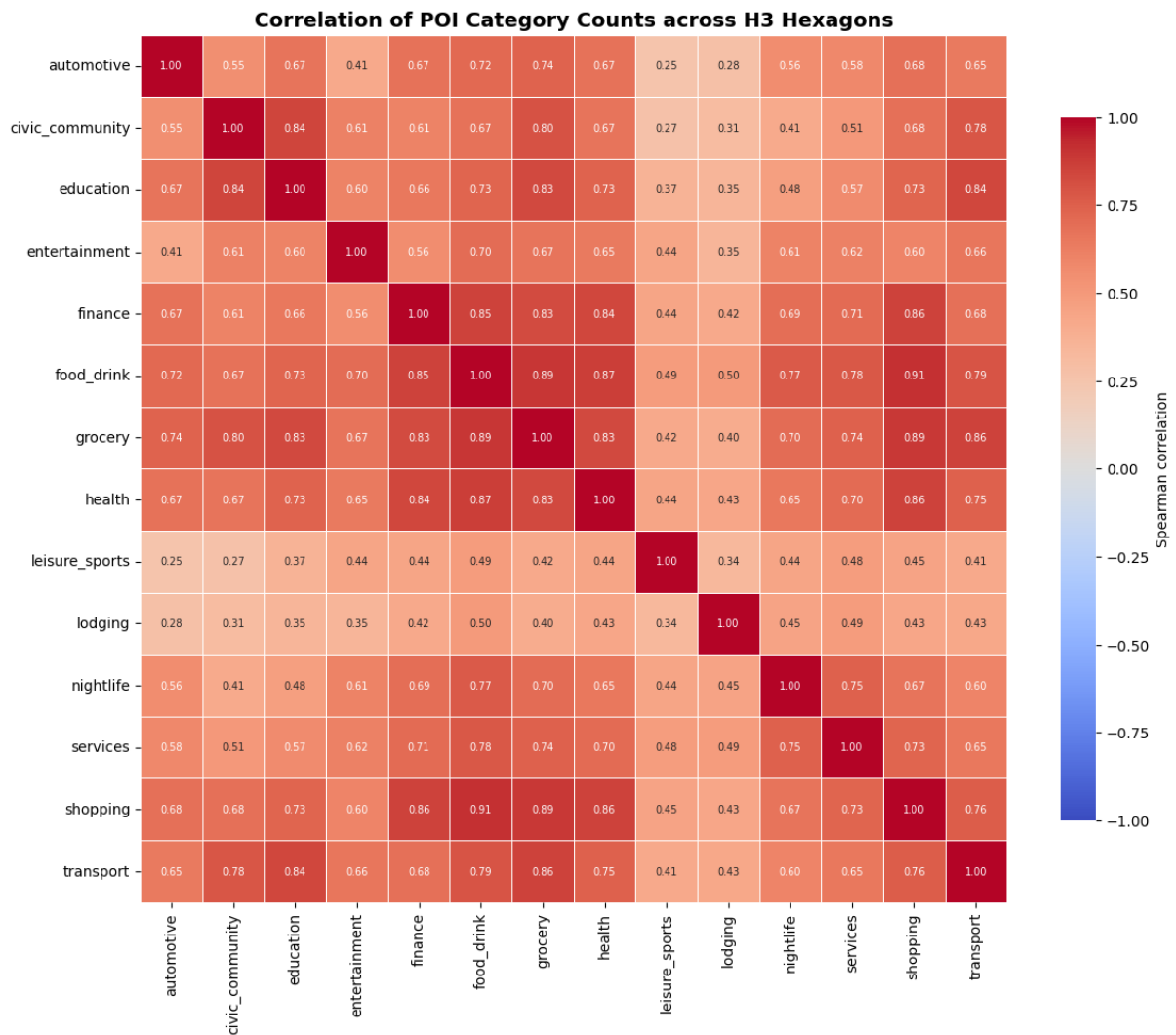


Figure 5: Spearman correlation of POI category counts across H3 hexagons (resolution 7).

A.3 Descriptive Analysis: Spatial (2.1)

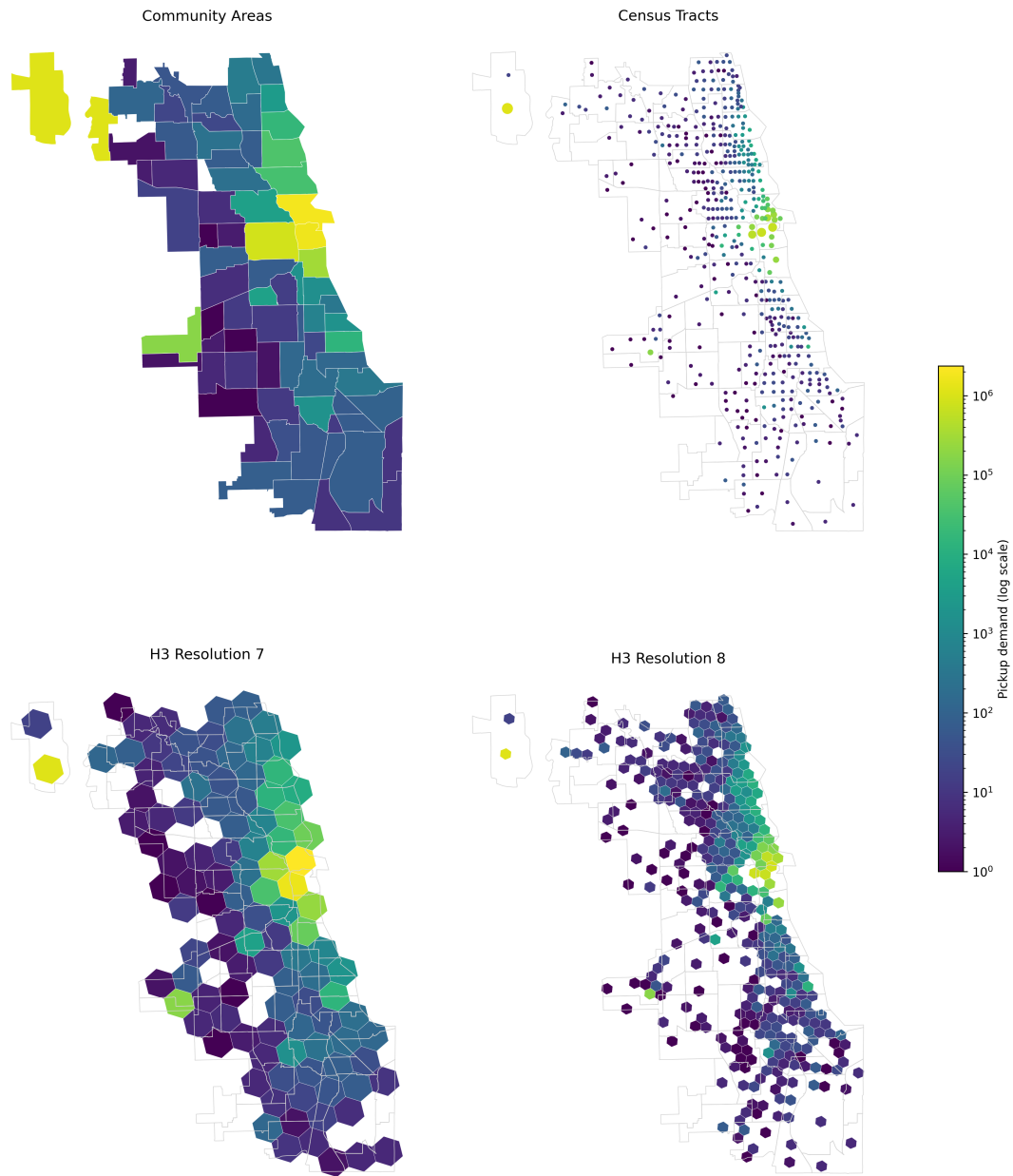


Figure 6: Spatial pickup demand distribution for different spatial entities.

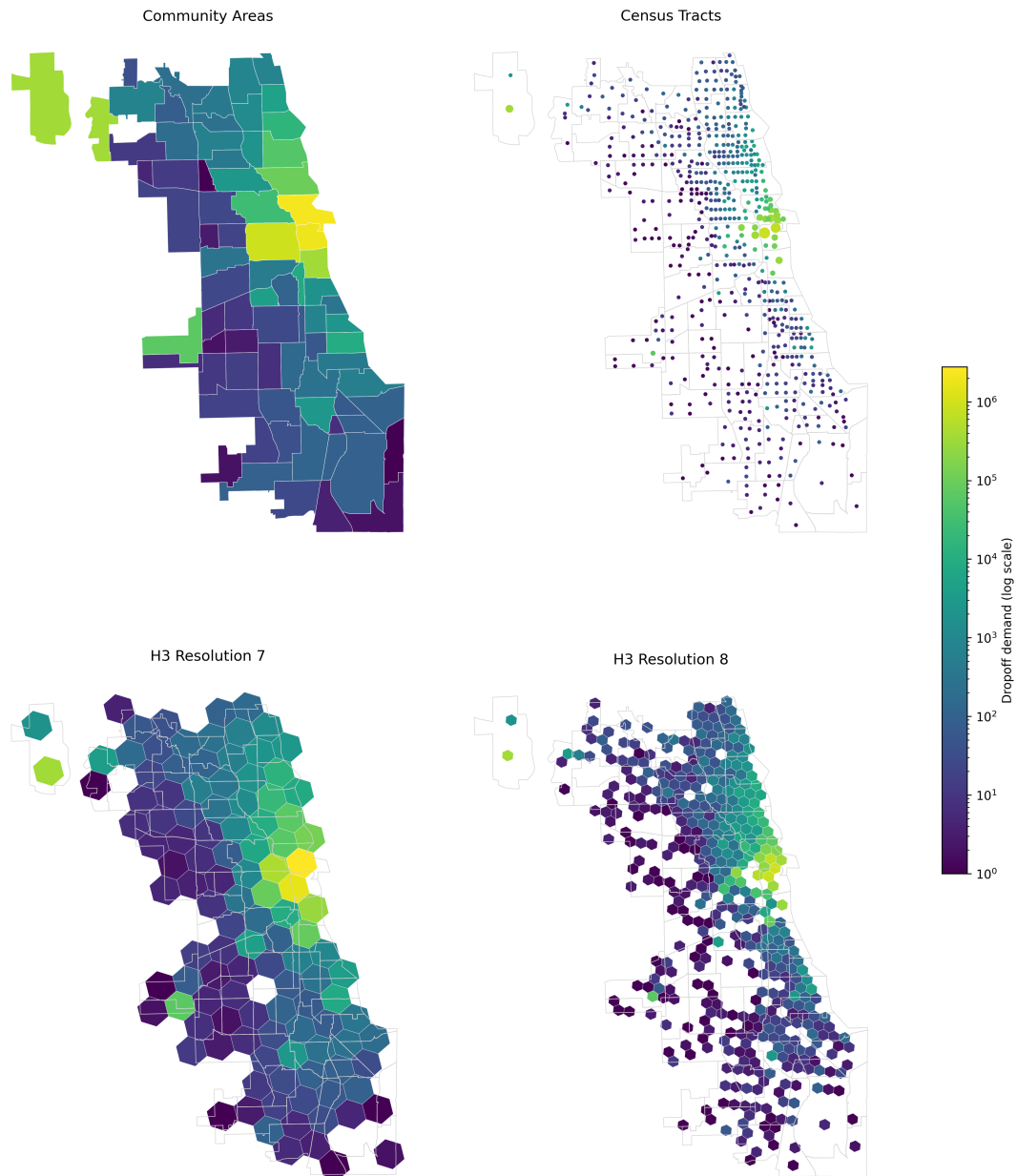


Figure 7: Spatial dropoff demand distribution for different spatial entities.

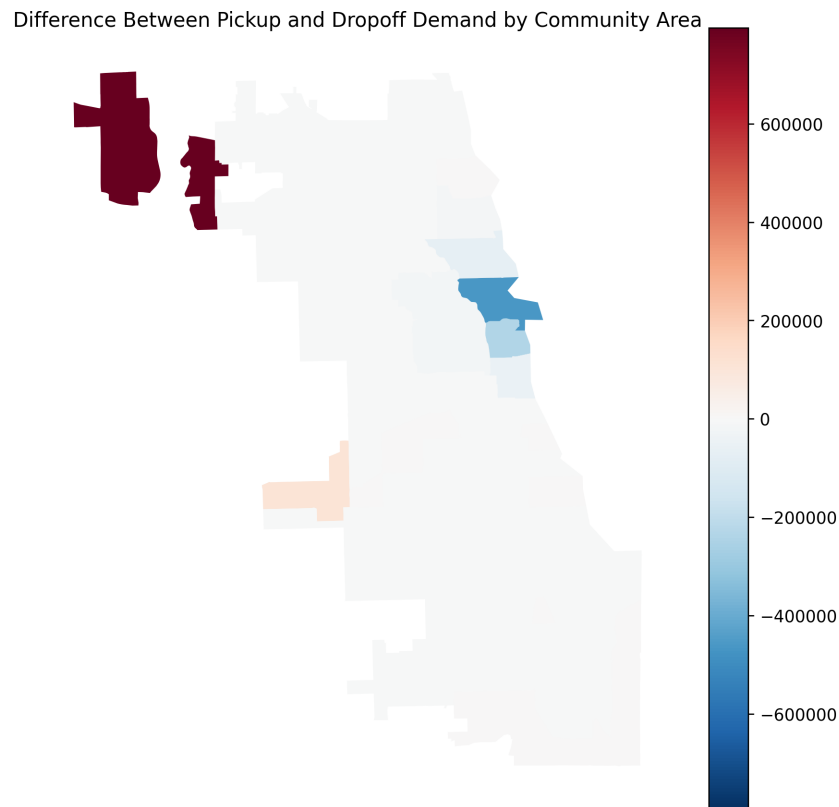


Figure 8: Difference between pickup and dropoff demand across community areas. Positive values indicate net pickup regions, while negative values indicate net dropoff regions.

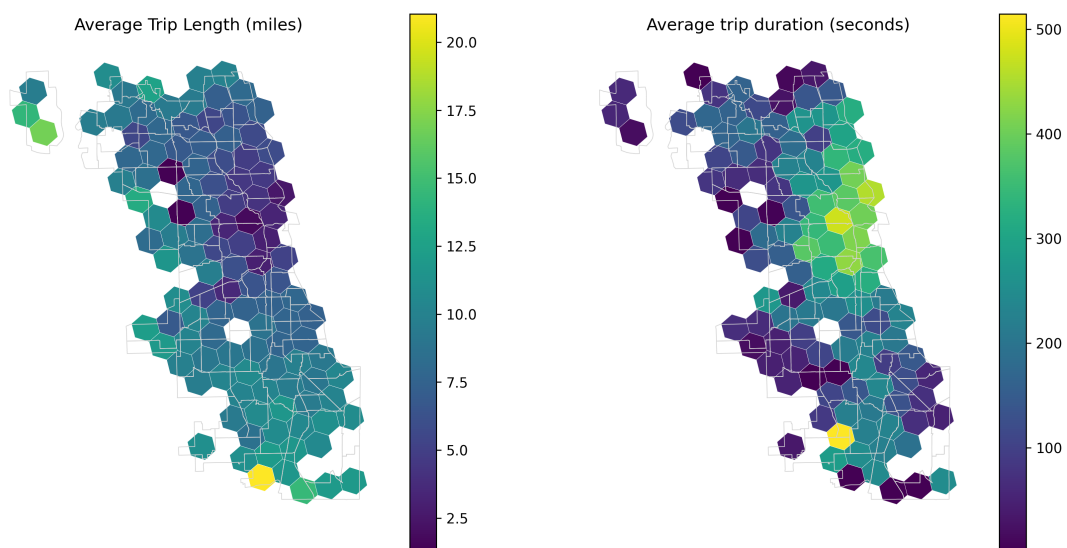


Figure 9: Average Trip Length and average Trip Duration.

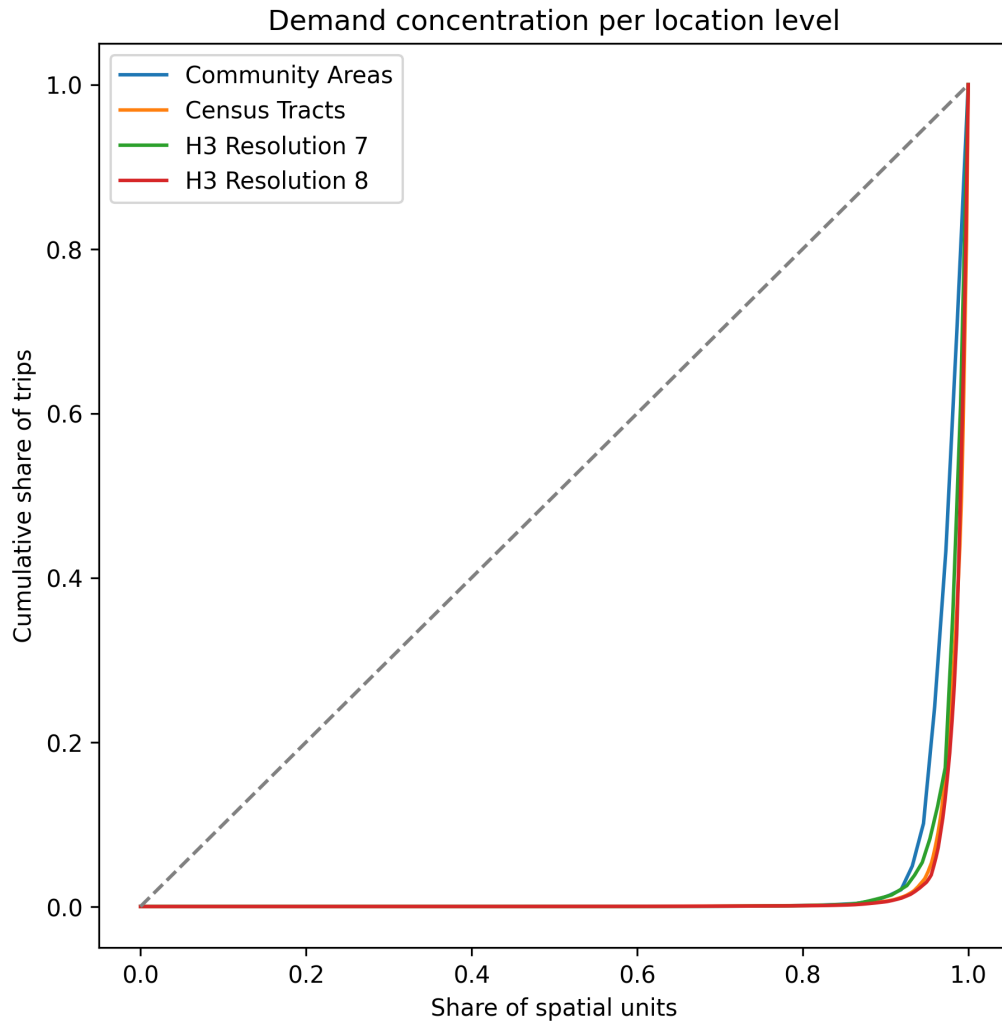


Figure 10: Demand concentration across different spatial resolutions measured by Lorenz curves. The increasing curvature with finer spatial resolutions indicates stronger concentration of taxi demand in a small number of spatial units.

Table 2: Spatial sparsity statistics across spatial resolutions. Increasing spatial resolution substantially increases the number of empty unit-hour cells, while H3 Resolution 7 provides a balance between spatial detail and data density.

Spatial resolution	Number of regions	Unit-hour cells	Non-empty cells	Empty cells (%)	Median active hours / region	Mean pickups / non-empty cell
Community Areas	77	1,594,824	874,335	45.2	11,399	14.81
Census Tracts	496	10,273,152	395,926	96.1	9	15.29
H3 Resolution 7	115	2,381,880	902,523	62.1	7,215	14.35
H3 Resolution 8	393	8,139,816	1,142,127	86.0	21	11.34
H3 Resolution 9	560	11,598,720	1,251,215	89.2	15	10.35

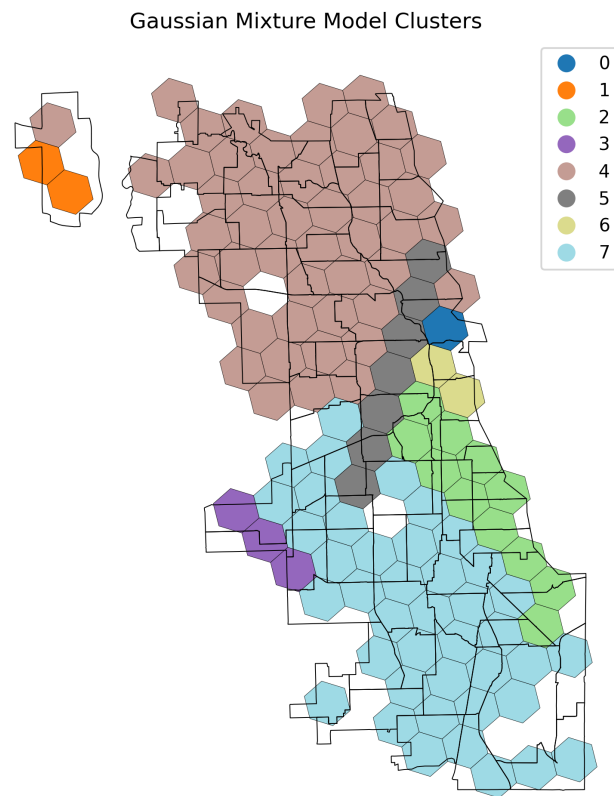


Figure 11: Gaussian Mixture Model (GMM) hot spots.

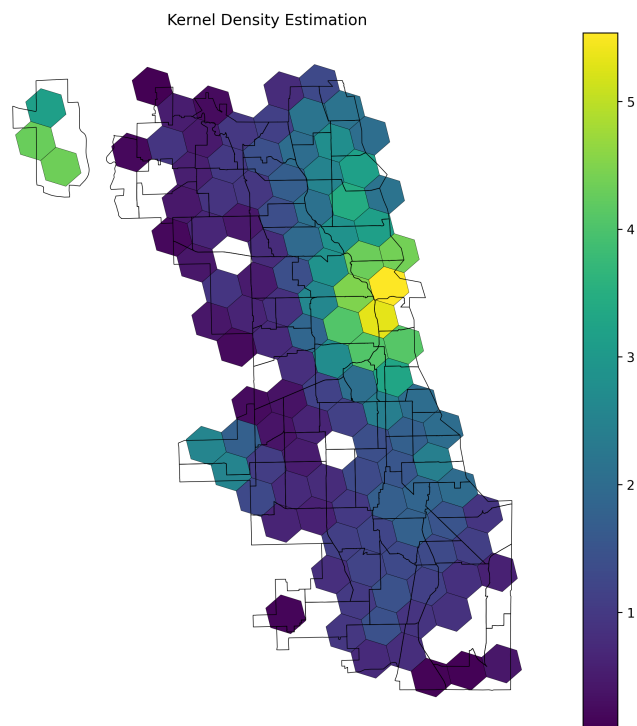


Figure 12: Kernel density estimation of taxi demand.

A.4 Descriptive Analysis: Temporal (2.2)

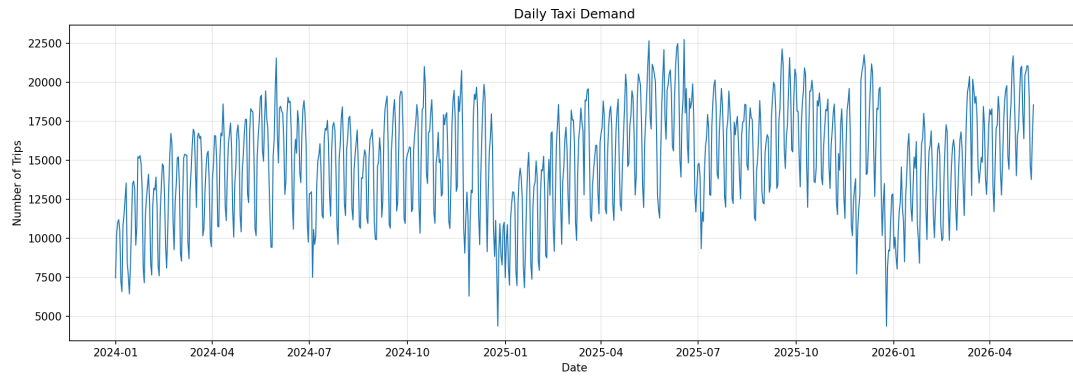


Figure 13: Daily taxi demand over the observation period.

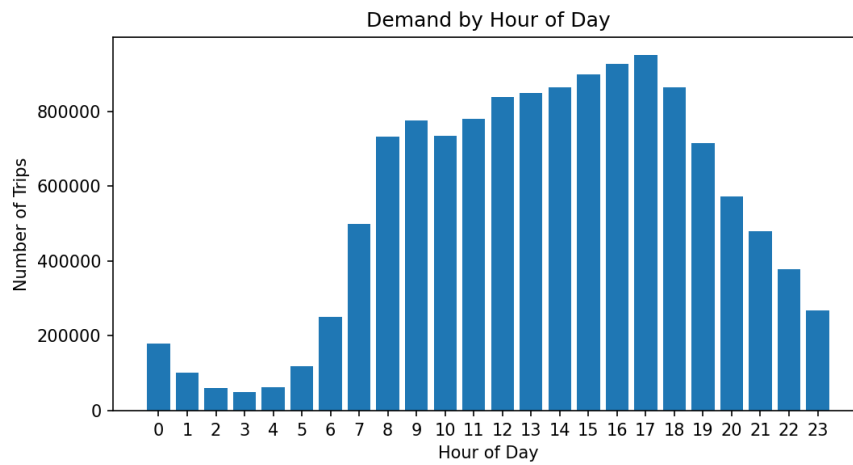


Figure 14: Hourly taxi demand.

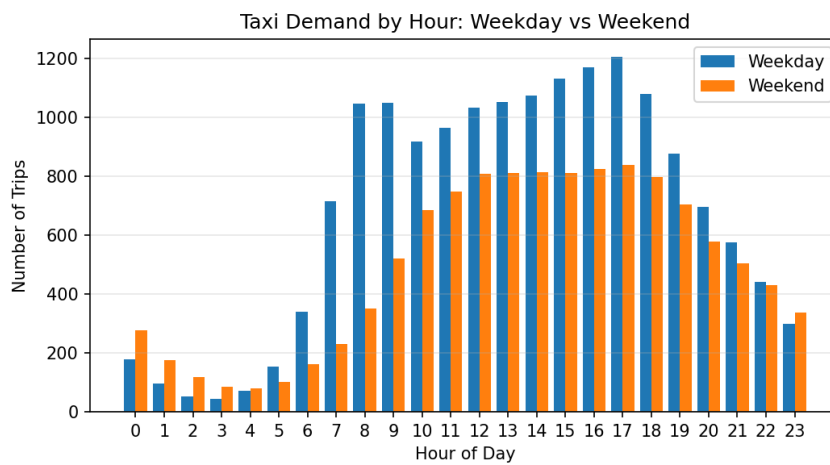


Figure 15: Hourly demand on weekdays and weekends.

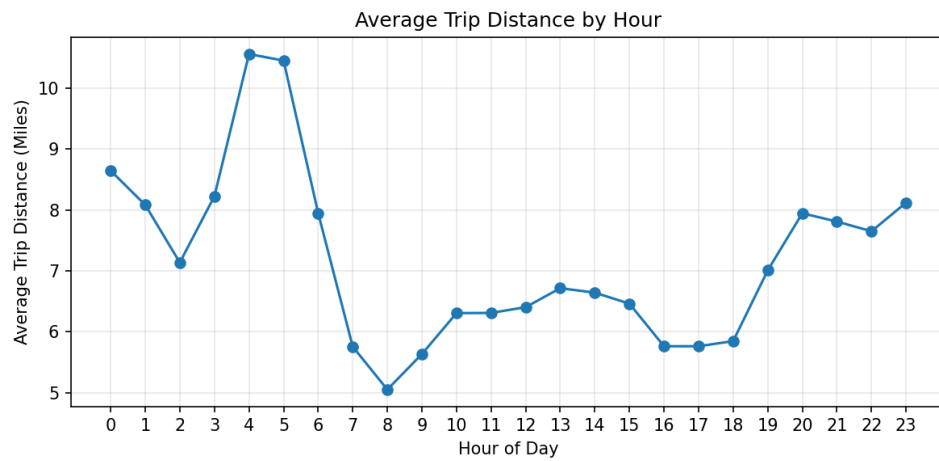


Figure 16: Average trip distance by hour.

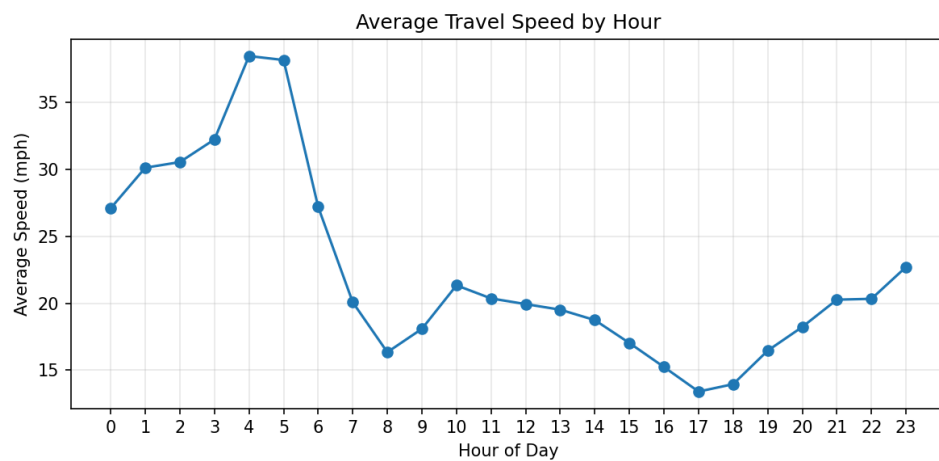


Figure 17: Average travel speed by hour.

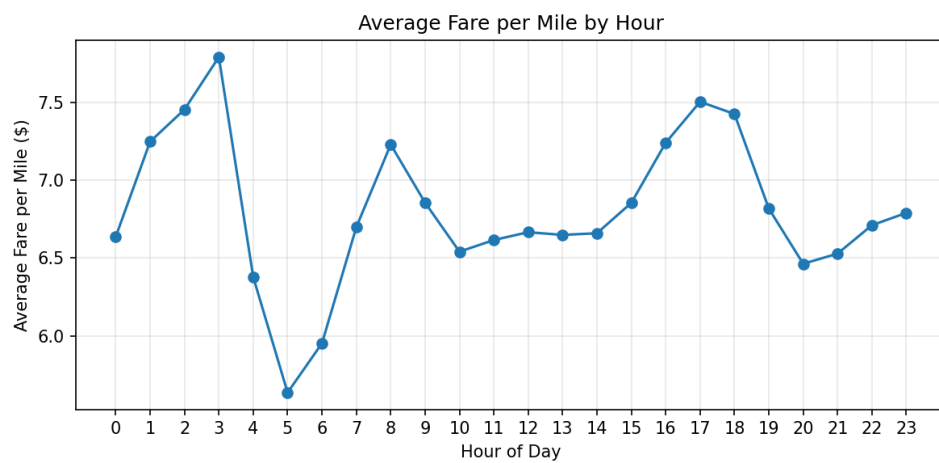


Figure 18: Average fare per mile by hour.

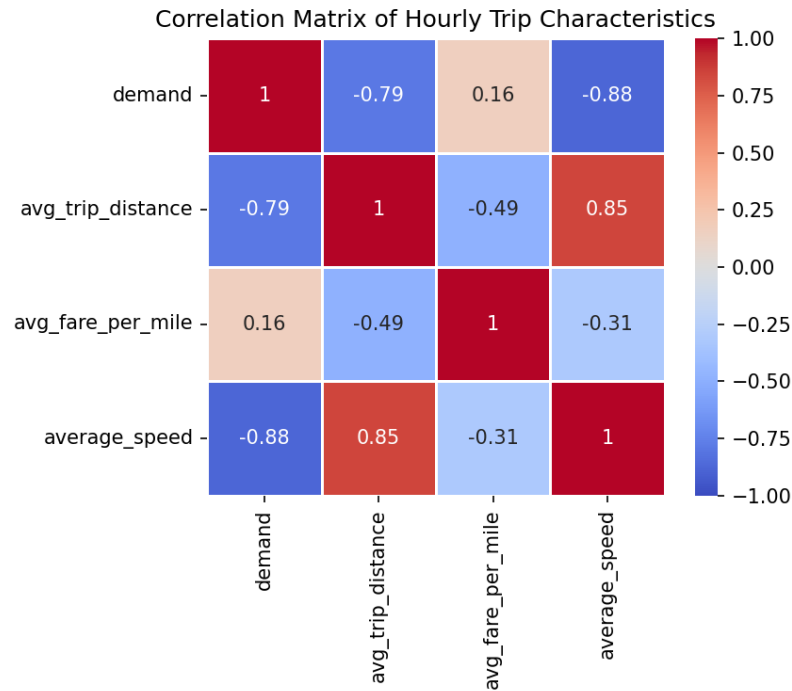


Figure 19: Correlation matrix of hourly demand and trip characteristics.

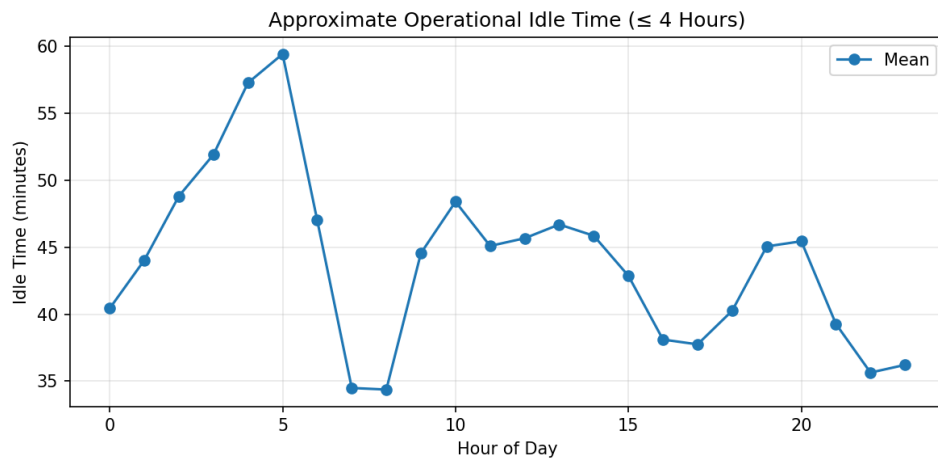


Figure 20: Approximate operational idle time by hour.

A.5 Feature Engineering (3.1)

Table 3: List of all Engineered Features.

Engineered Feature	Description	DataType
distance_to_loop	Great-circle (haversine) distance from each grid cell's coordinates to the Chicago Loop (downtown reference point), used as a proxy for centrality/urban density.	float64
wind_speed	Magnitude of the horizontal wind vector, computed as $\sqrt{u_{10}^2 + v_{10}^2}$ from the 10m u/v wind components.	float64
wind_dir	Meteorological wind direction (direction the wind blows from), computed via $\text{atan2}(u_{10}, v_{10})$ and converted to compass degrees (0-360°).	float64

Table 3: List of all Engineered Features.

Engineered Feature	Description	Data Type
is_holiday	Binary indicator flagging official U.S. federal holidays observed in Illinois, expected to shift ride-hailing demand relative to regular weekdays.	int64
is_near_holiday	Binary indicator flagging the calendar day immediately before or after a public holiday, capturing anticipatory or spillover demand effects.	int64
hour_of_day	Hour extracted from the timestamp, capturing intraday demand cycles. Only present in hourly-resolution grids.	int64
hour_sin	Sine transformation of hour_of_day, encoding time of day on a continuous circular scale. Only present in hourly-resolution grids.	float64
hour_cos	Cosine transformation of hour_of_day, paired with hour_sin to avoid the discontinuity between hour 23 and hour 0. Only present in hourly-resolution grids.	float64
month	Calendar month extracted from the timestamp, capturing seasonal variation.	int64
month_sin	Sine transformation of month, encoding the annual cycle continuously.	float64
month_cos	Cosine transformation of month, paired with month_sin to avoid the discontinuity between December and January.	float64
day_of_week	Numeric day of week extracted from the timestamp (0 = Monday, ..., 6 = Sunday), capturing weekly demand patterns.	int64
is_weekend	Binary indicator flagging Saturday and Sunday (day_of_week $\geq$ 5), expected to correlate with different ride purposes (leisure vs. commute).	int64
weekday	Numeric day of week extracted from the timestamp; identical in definition to day_of_week (0 = Monday, ..., 6 = Sunday).	int64
season	Meteorological season derived from month, encoded as an integer (1 = Winter, 2 = Spring, 3 = Summer, 4 = Autumn), capturing broader seasonal demand and weather regimes.	int64
season_sin	Sine transformation of season, representing the seasonal cycle continuously.	float64
season_cos	Cosine transformation of season, paired with season_sin to avoid a discontinuity between the last and first season of the year.	float64
base_demand	Historical average of Total_Trip_Start, computed on the training split only and grouped by spatial cell, month, day_of_week, and (for hourly grids) hour_of_day; unseen key combinations are filled with the global training mean.	float64

A.6 SVR Hyperparameter Search Grid (3.2)

Both SVR notebooks share the identical two-stage search defined in 3.2 `PredictionCommonGround.ipynb`, so the direct and residual results stay directly comparable. The grid search runs only for the kernel that wins the scan, not for all three.

Table 4: Two-stage SVR hyperparameter search grid, shared by the direct (3.2a) and residual (3.2b) models.

Stage	Setting	Values searched
Kernel scan	Kernels	linear , poly , rbf
	Held fixed during the scan	C = 1, gamma = "scale"
	Selection criterion	Validation R ² in count space
	Training sample	4,000 rows per class (8,000 total)
Grid search	C	1, 10, 50
	gamma	"scale", 0.05, 0.2
	degree (polynomial kernel only)	2, 3
	Cross-validation	3-fold
	Selection criterion	Count-space MAE
	Training sample	8,000 rows per class (16,000 total)

Stage	Setting	Values searched
Feature sets	Candidates	basic , basic+poi , basic+weather , basic+poi+weather
	Selection criterion	Validation R^2 in count space
Scoring	Validation subsample	50,000 rows; the level's winner is re-scored on the full validation split

The four candidate feature sets are built from the same blocks at every level. **basic** is hour (hourly grids only) plus calendar, location and history; the other three add the POI and weather blocks to it.

Table 5: Feature blocks composing the four candidate sets. Resulting widths are 10 / 24 / 15 / 29 features on daily grids and 12 / 26 / 17 / 31 on hourly grids, for **basic** / **basic+poi** / **basic+weather** / **basic+poi+weather** respectively.

Block	Features	Count
Hour (hourly grids only)	hour_sin, hour_cos	2
Calendar	month_sin, month_cos, is_weekend, is_holiday, is_near_holiday, day_of_week	6
Location	lat, lon, distance_to_loop	3
History	base_demand	1
POI	14 poi_cat_* category counts	14
Weather	2m_temp_c, total_precip_mm, snow_cov, snow_depth, wind_speed	5

A.7 SVR: Direct Model (3.2a)

Table 6: Winning SVR per level, with skill against the historical-mean baseline.

	spatial	temporal	feature_set	kernel	base_MAE	val_MAE	val_RMSE	val_R2	skill_MAE	skill_RMSE
0	h3_7	daily	basic+poi	rbf	15.8295	15.7997	88.8310	0.9494	0.0019	0.0069
1	community	daily	basic+poi	rbf	36.1879	37.4429	139.8964	0.9578	-0.0347	-0.0769
2	census	daily	basic+poi	rbf	3.8284	4.0758	27.3988	0.9341	-0.0646	-0.0354
3	community	hourly	basic+poi	rbf	2.5250	2.7382	10.3958	0.9133	-0.0844	-0.1556
4	h3_7	hourly	basic	rbf	0.8861	0.9858	6.2626	0.9148	-0.1125	-0.0348
5	census	hourly	basic+poi+weather	rbf	0.2438	0.3302	2.0741	0.8795	-0.3547	-0.0468

A.8 SVR: Residual-on-Baseline Model (3.2b)

Table 7: Winning residual-on-baseline SVR per level, with skill against the historical-mean baseline.

	spatial	temporal	feature_set	kernel	base_MAE	val_MAE	val_RMSE	val_R2	skill_MAE	skill_RMSE
0	community	daily	basic+weather	linear	36.1879	34.2541	118.3257	0.9698	0.0534	0.0891
1	h3_7	daily	basic+poi	linear	15.8295	15.6367	84.9372	0.9538	0.0122	0.0504
2	census	daily	basic+poi+weather	rbf	3.8284	3.8174	26.1691	0.9399	0.0029	0.0111
3	h3_7	hourly	basic+weather	linear	0.8861	0.8838	5.5285	0.9336	0.0026	0.0865
4	community	hourly	basic+poi+weather	rbf	2.5250	2.5498	9.2293	0.9316	-0.0098	-0.0259
5	census	hourly	basic	linear	0.2438	0.3221	1.9674	0.8915	-0.3214	0.0071

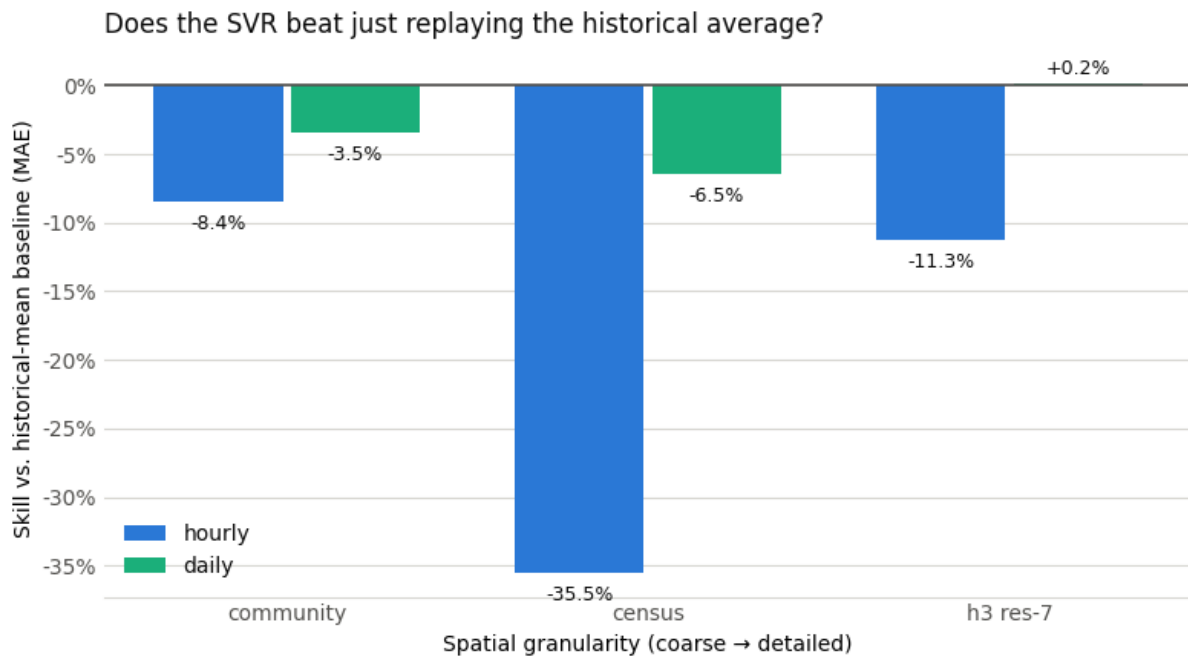


Figure 21: SVR skill against the historical-mean baseline, by spatial and temporal granularity.

Feature importance: reliance vs. necessity (= block not in this model's feature set)

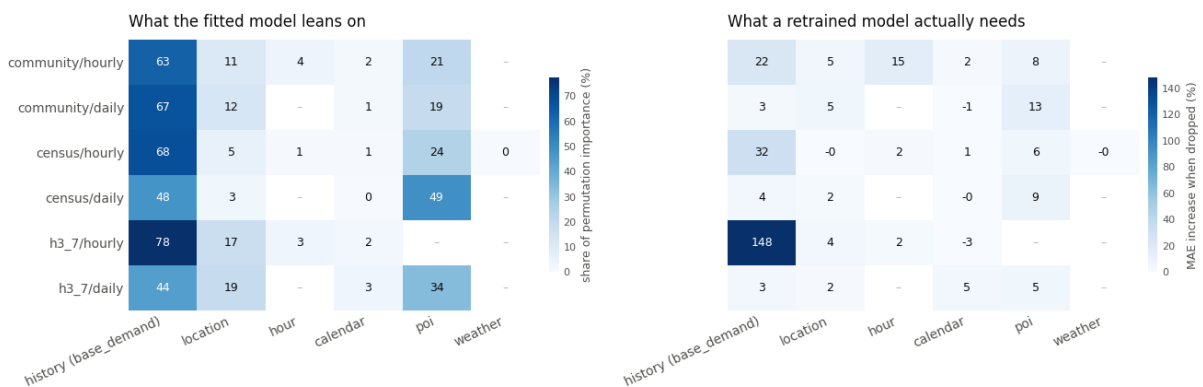


Figure 22: Left: what the fitted model leans on (grouped permutation importance, as each block's share of the level's total). Right: what a retrained model actually needs (drop-column ablation, % increase in MAE when the block is removed). The two disagree sharply on base_demand — the fitted model depends on it almost entirely, yet a model refitted without it loses very little, because lat/lon/hour/day-of-week can reconstruct it.

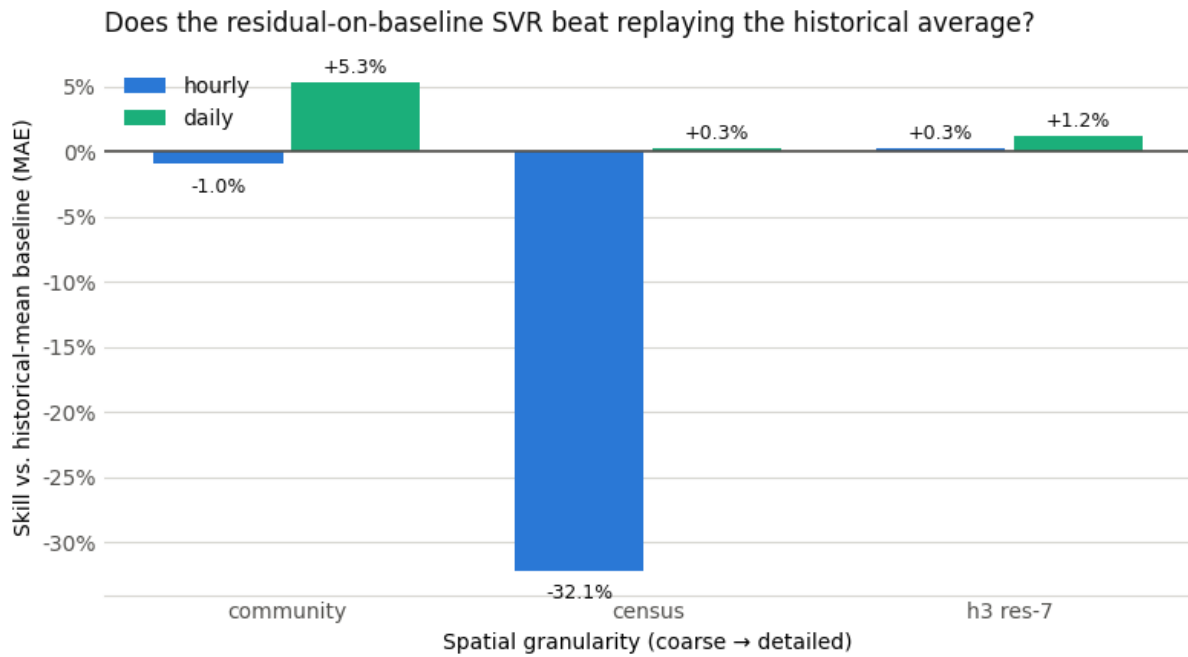


Figure 23: Residual-on-baseline SVR skill against the historical-mean baseline, by spatial and temporal granularity.

Table 8: Direct SVR (3.2a) vs. residual-on-baseline SVR trained from scratch (this notebook), per level, on the validation split.

spatial	temporal	feature_set	kernel	skill_MAE_direct	skill_MAE_residual	delta_skill_MAE
0 h3_7	hourly	basic+weather	linear	-0.1125	0.0026	0.1151
1 community	daily	basic+weather	linear	-0.0347	0.0534	0.0881
2 community	hourly	basic+poi+weather	rbf	-0.0844	-0.0098	0.0746
3 census	daily	basic+poi+weather	rbf	-0.0646	0.0029	0.0675
4 census	hourly	basic	linear	-0.3547	-0.3214	0.0333
5 h3_7	daily	basic+poi	linear	0.0019	0.0122	0.0103

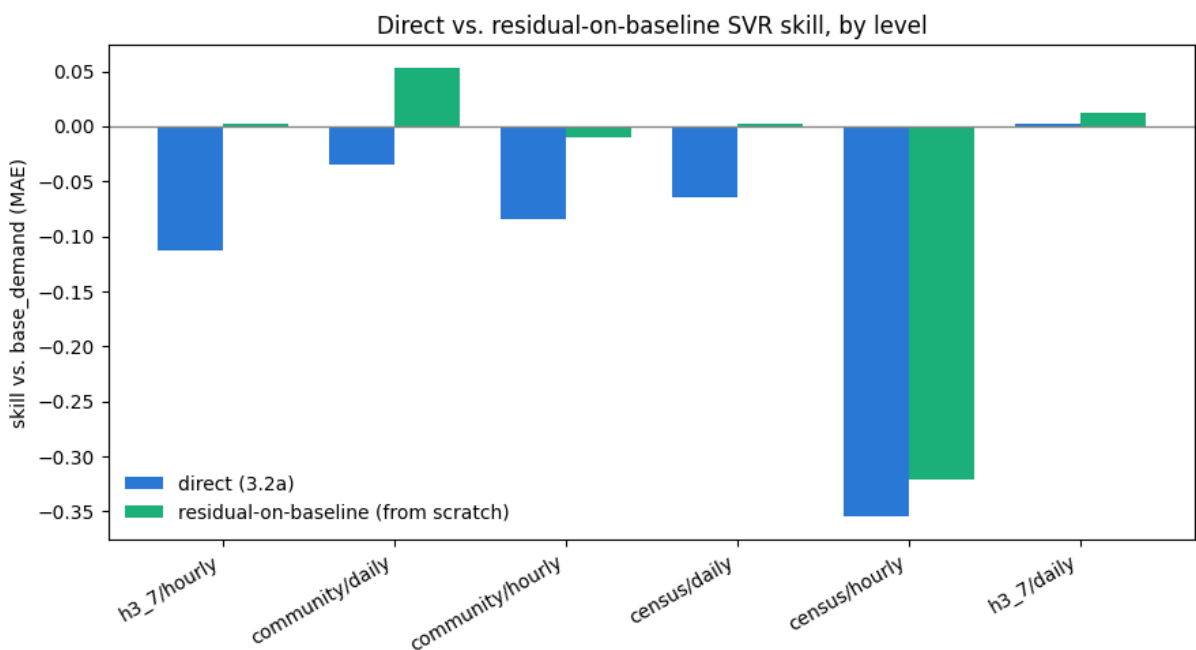


Figure 24: Direct (3.2a) vs. residual-on-baseline (this notebook) SVR skill, both independently tuned, per level.

A.9 Deep Learning Prediction (3.3)

The grid search was run independently for each of the six spatial and temporal combinations, with the winning configuration selected by validation MAE in count space.

Table 9: FNN hyperparameter search grid.

Stage	Setting	Values	Rationale
Architecture search	Hidden Layer configurations	[64,32,16], [128,64,32], [128,64,32,16], [256,128,64,32]	Pyramid configurations outperformed equal-width configurations in initial experiments.
	Learning rates	0.01, 0.001, 0.0001	Covers three orders of magnitude to account for level-specific convergence behaviour
	Dropout rates	0.0, 0.2	Higher rates consistently hurt performance in initial tests
	Configurations per level Selection criterion	24 (full grid) Validation MAE in count space	
Training	Optimizer	AdamW (weight decay 1e-4)	Decouples weight decay from the gradient update, avoiding L2 scaling issues with adaptive optimizers (Loshchilov & Hutter, 2019).
	Loss function	MSE	
	Batch size	250	
	Maximum epochs	15	
	Early stopping patience	3 epochs	Stops training after three consecutive epochs without validation improvement
Feature set	Candidates	basic+poi+weather (fixed)	
	Hourly features	31 total	
	Daily features	29 total	

Table 10: Winning FNN architecture per level, selected by validation MAE in count space over a grid of four layer configurations, three learning rates, and two dropout rates (24 configurations per level).

Level	Layers	Learning rate	Dropout
community / hourly	256, 128, 64, 32	0.001	0.0
community / daily	128, 64, 32, 16	0.01	0.0
h3_7 / hourly	128, 64, 32, 16	0.0001	0.0
h3_7 / daily	128, 64, 32, 16	0.01	0.2
census / hourly	64, 32, 16	0.001	0.0
census / daily	64, 32, 16	0.001	0.2

Table 11: FNN test performance per level against the historical-mean baseline ($\text{skill} = 1 - \text{MAE_model} / \text{MAE_baseline}$), sorted by skill (MAE).

Level	MAE	RMSE	R ²	Skill (MAE)	Skill (RMSE)
community / hourly	2.620	9.678	0.925	+9.7%	+12.7%
h3_7 / hourly	1.041	7.358	0.883	+2.5%	+1.0%
census / hourly	0.276	2.218	0.844	-1.0%	-6.1%
community / daily	54.518	190.617	0.921	-10.0%	-4.8%
census / daily	6.355	42.094	0.819	-29.2%	-42.7%

Level	MAE	RMSE	R ²	Skill (MAE)	Skill (RMSE)
h3_7 / daily	28.209	155.722	0.844	-31.9%	-28.7%

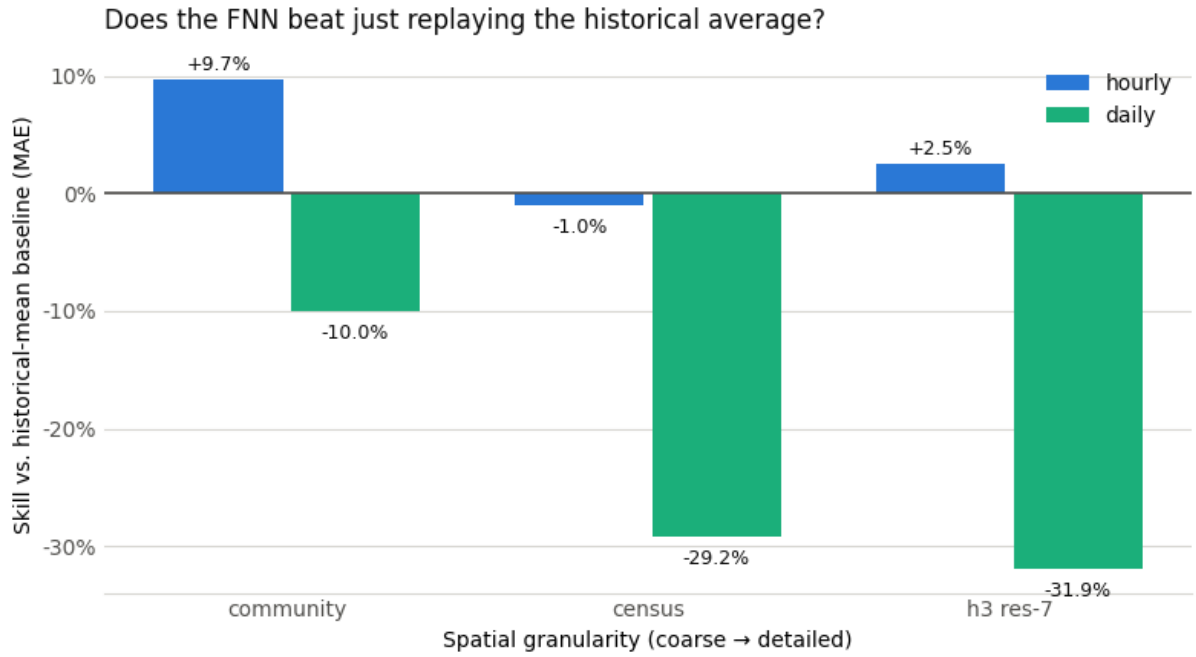


Figure 25: FNN skill against the historical-mean baseline, by spatial and temporal granularity.

A.10 SVM vs. Neural Network Performance Comparison (3.4)

Table 12: Overall TEST performance at community level.

(a) community/daily			
	MAE	RMSE	R2
Model			
Baseline	49.5750	181.9158	0.9283
Best SVM (Residual SVR)	46.3370	164.6925	0.9413
Neural Network (FNN)	54.5184	190.6171	0.9213
(b) community/hourly			
	MAE	RMSE	R2
Model			
Baseline	2.9029	11.0871	0.9019
Best SVM (Residual SVR)	2.8777	11.2611	0.8988
Neural Network (FNN)	2.6201	9.6776	0.9252

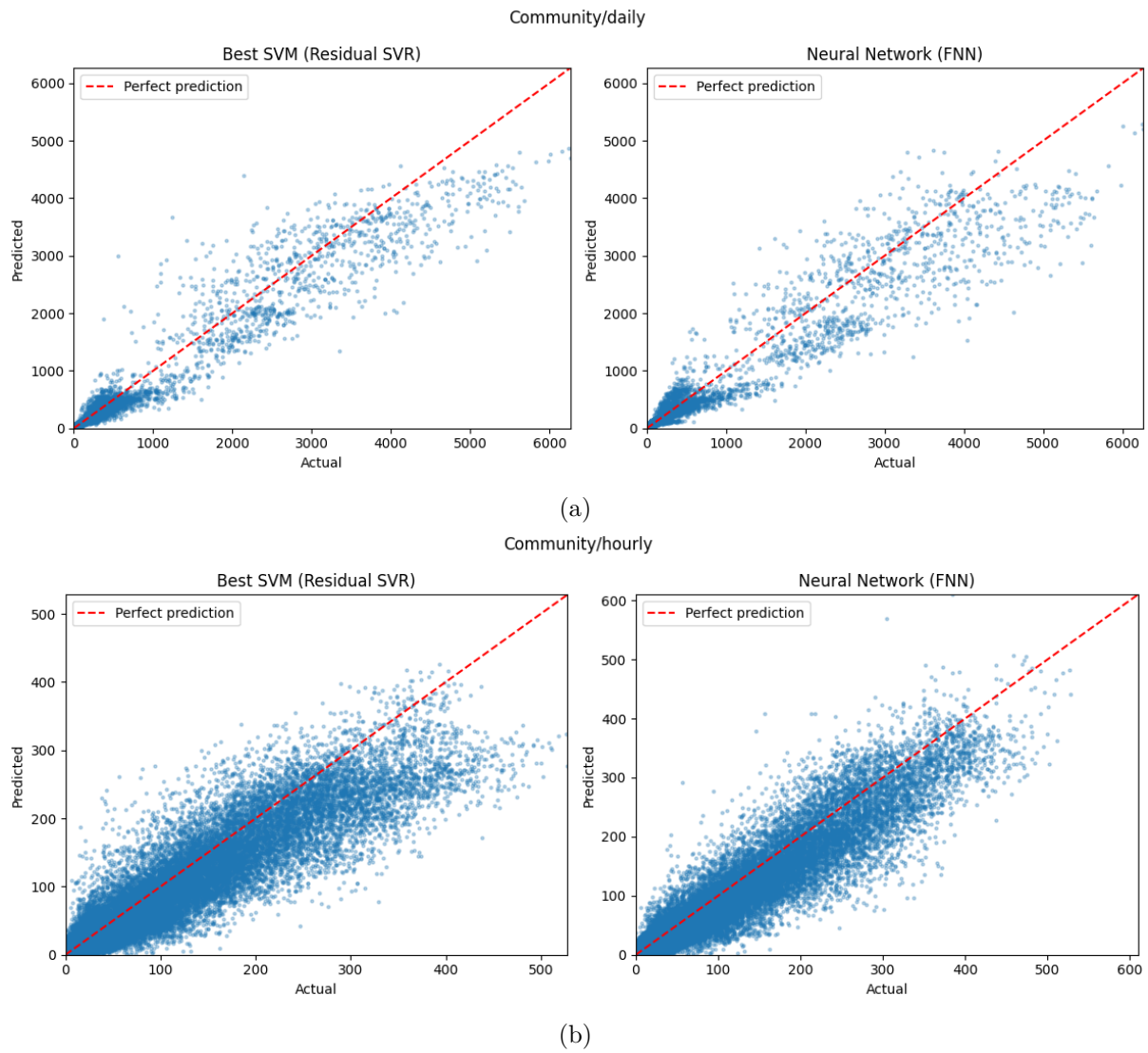
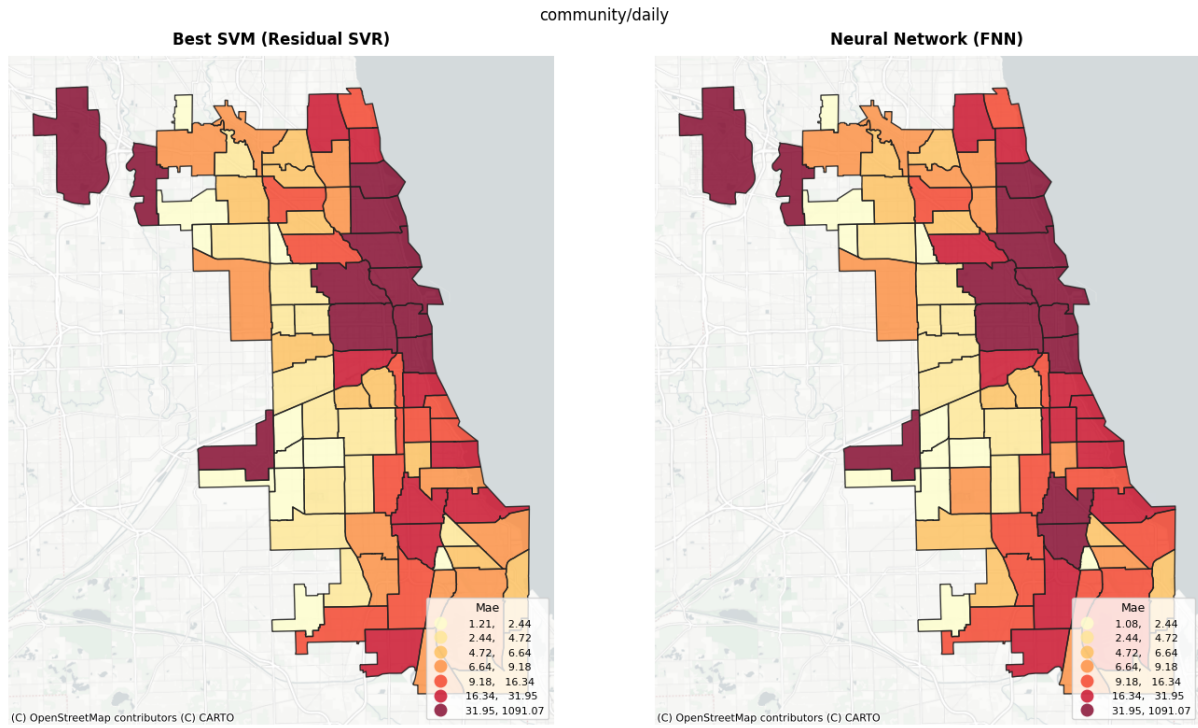
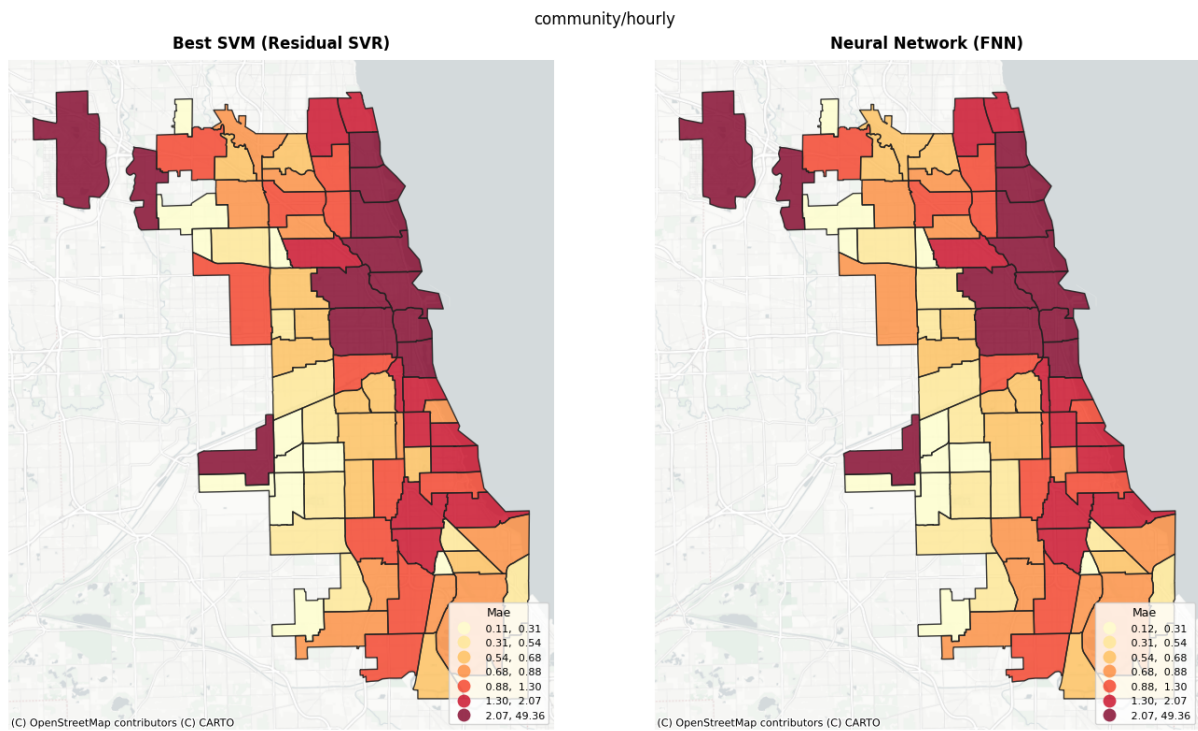


Figure 26: Actual vs. predicted demand per temporal resolution.

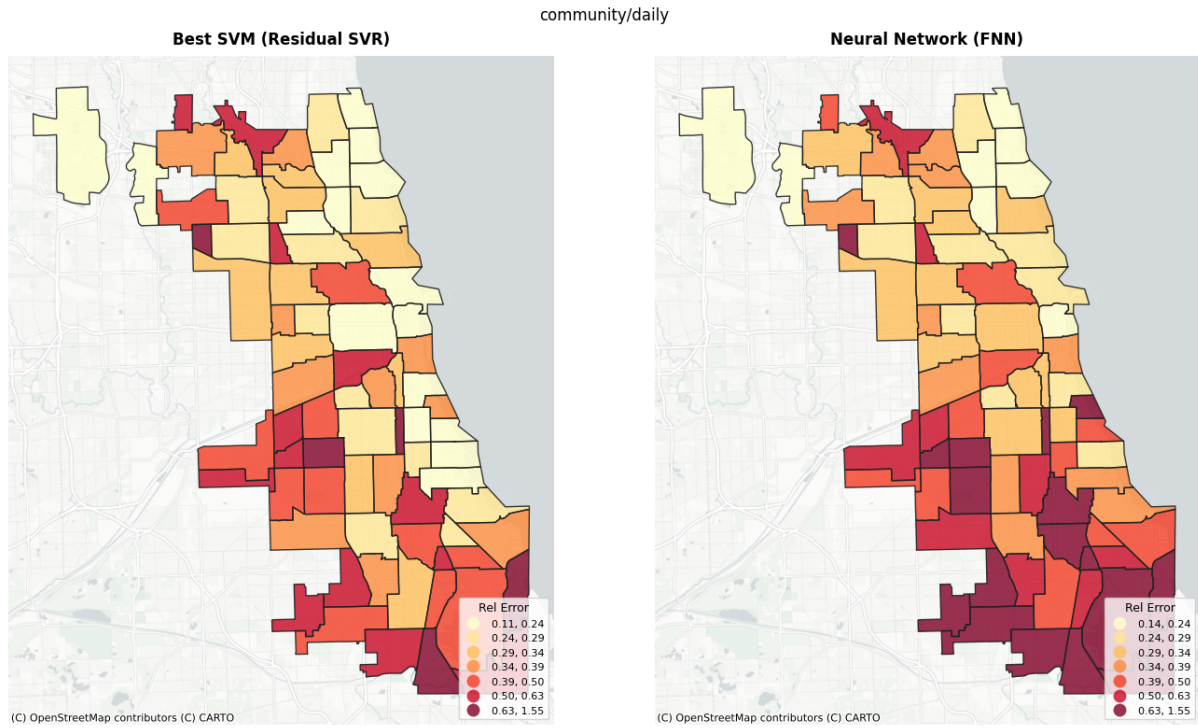


(a)



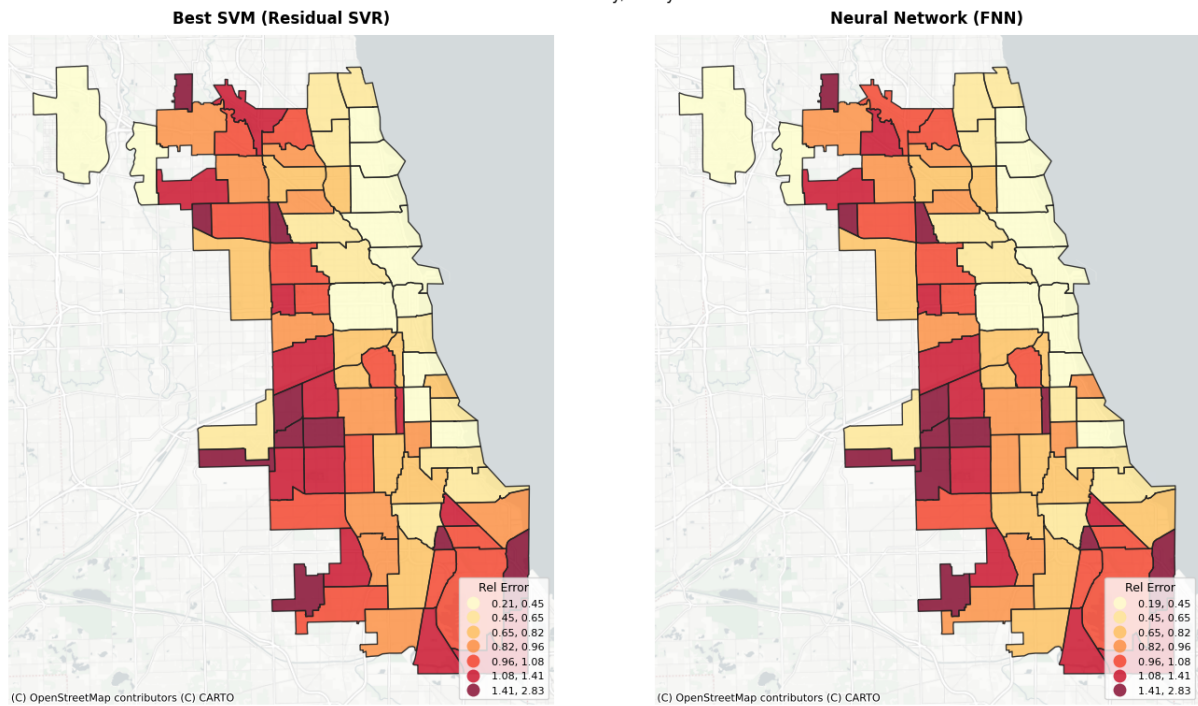
(b)

Figure 27: MAE per community area per temporal resolution.



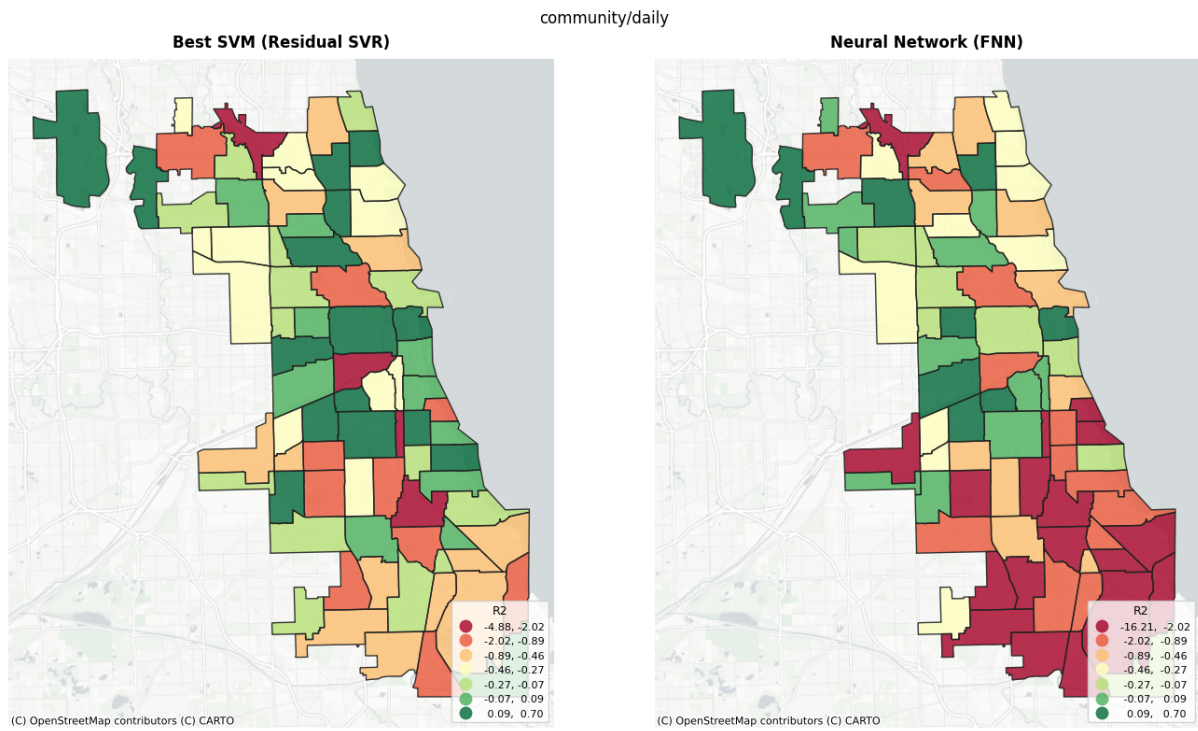
(a)

community/hourly

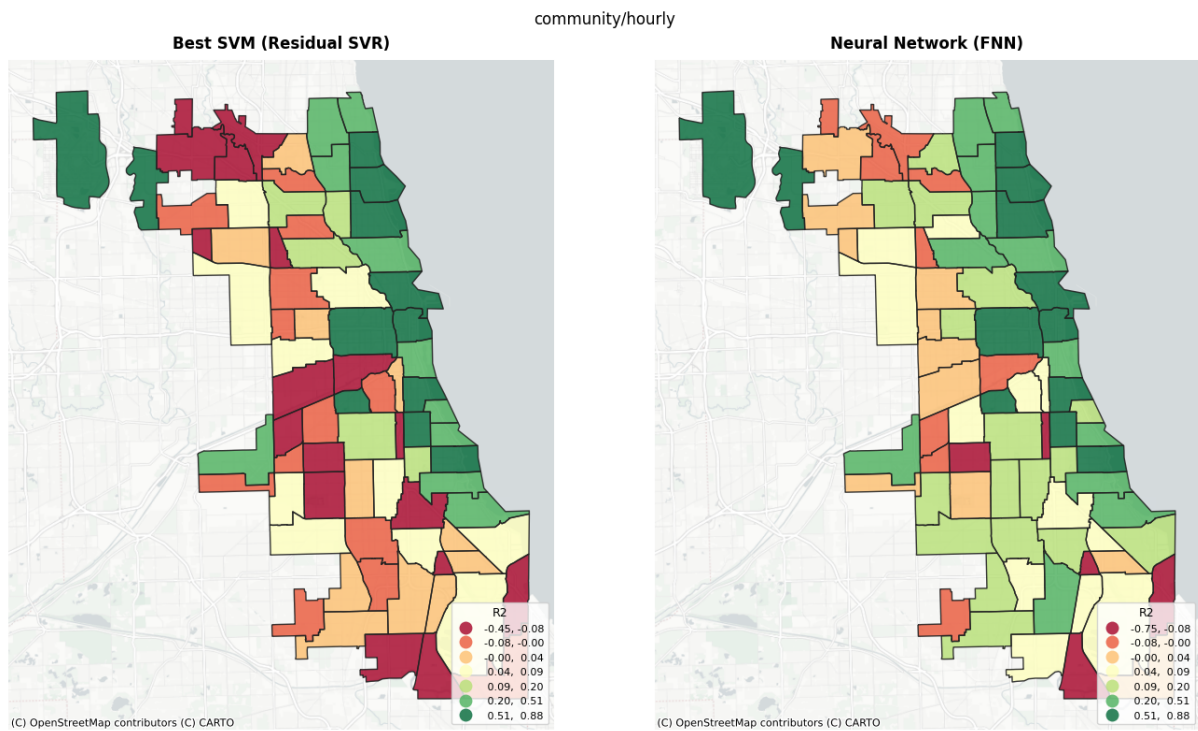


(b)

Figure 28: Relative error per community area per temporal resolution.

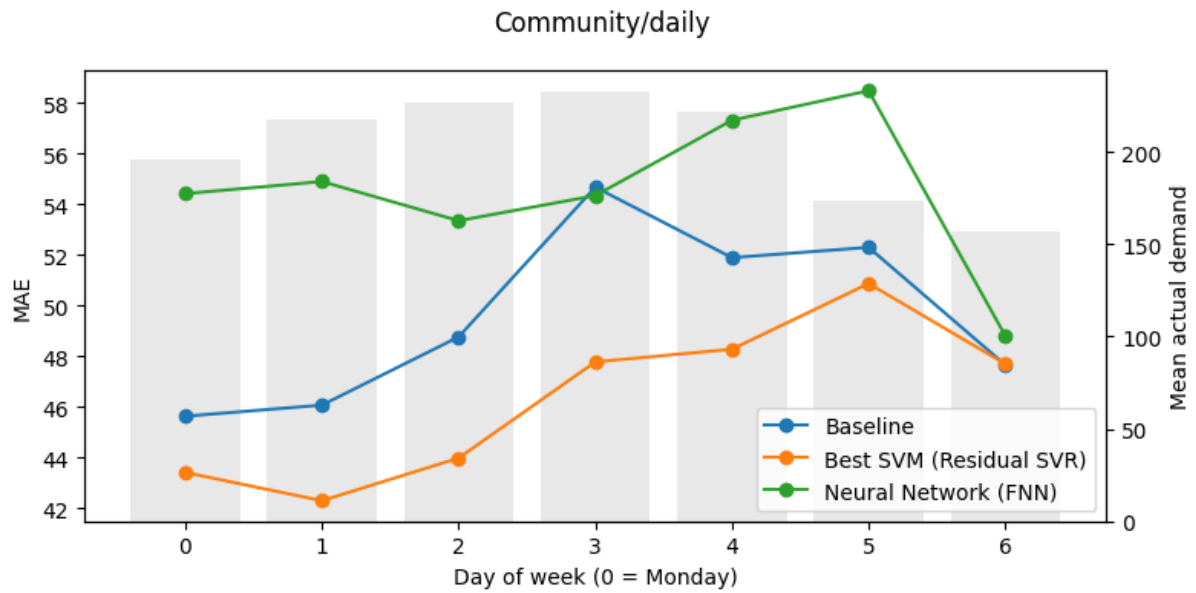


(a)

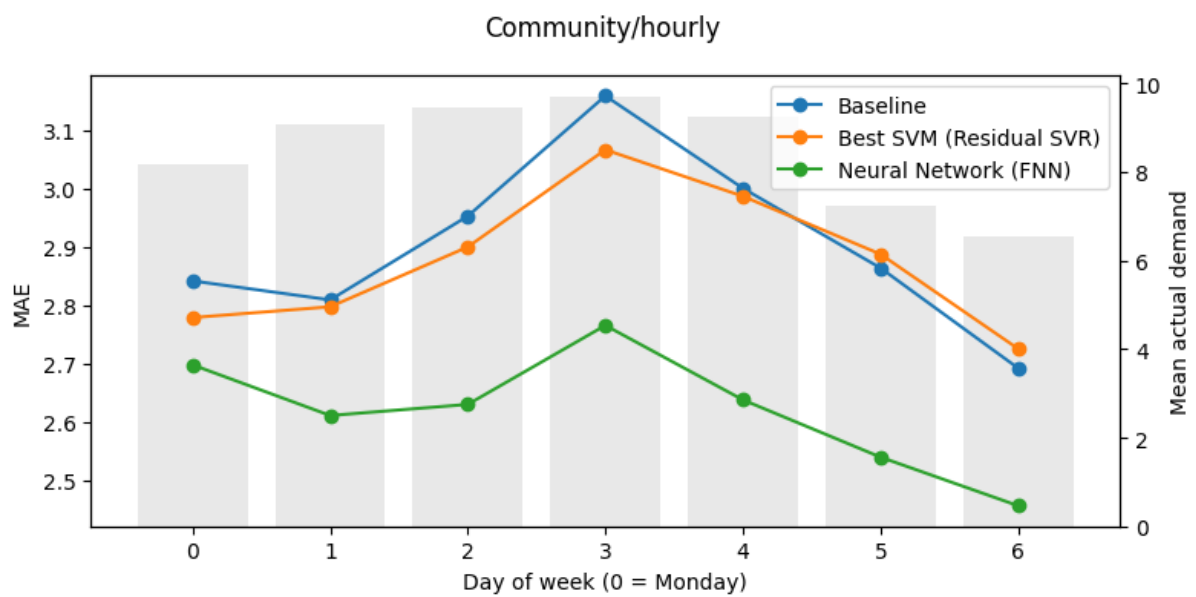


(b)

Figure 29: R^2 per community area per temporal resolution.

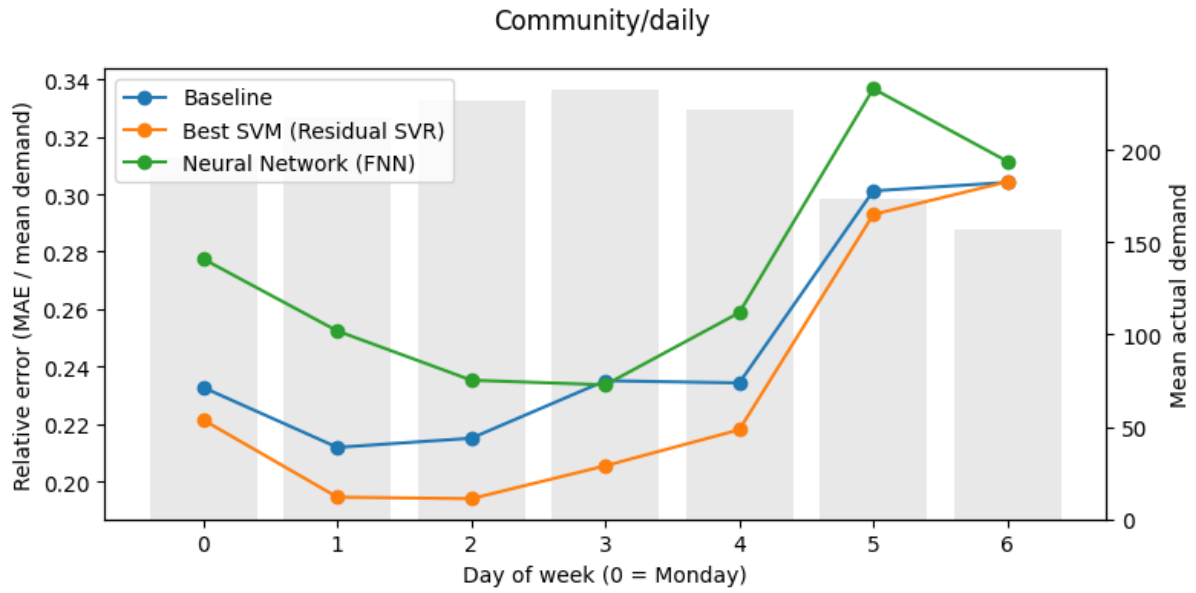


(a)

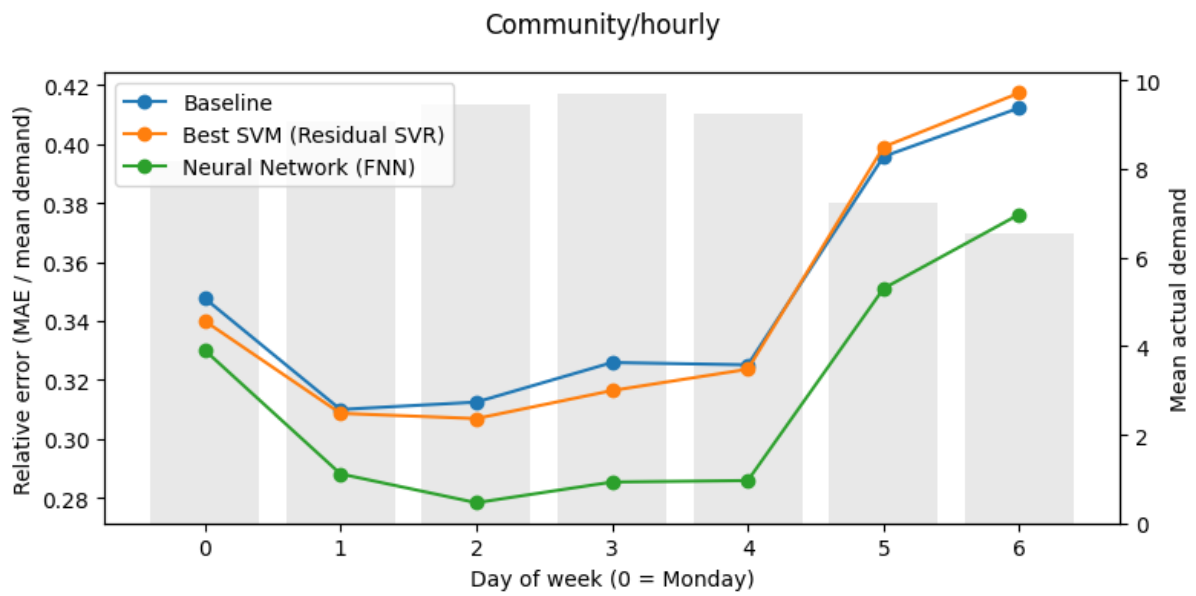


(b)

Figure 30: Test MAE by day of week per temporal resolution.



(a)



(b)

Figure 31: Test relative error (MAE / mean demand) by day of week per temporal resolution.

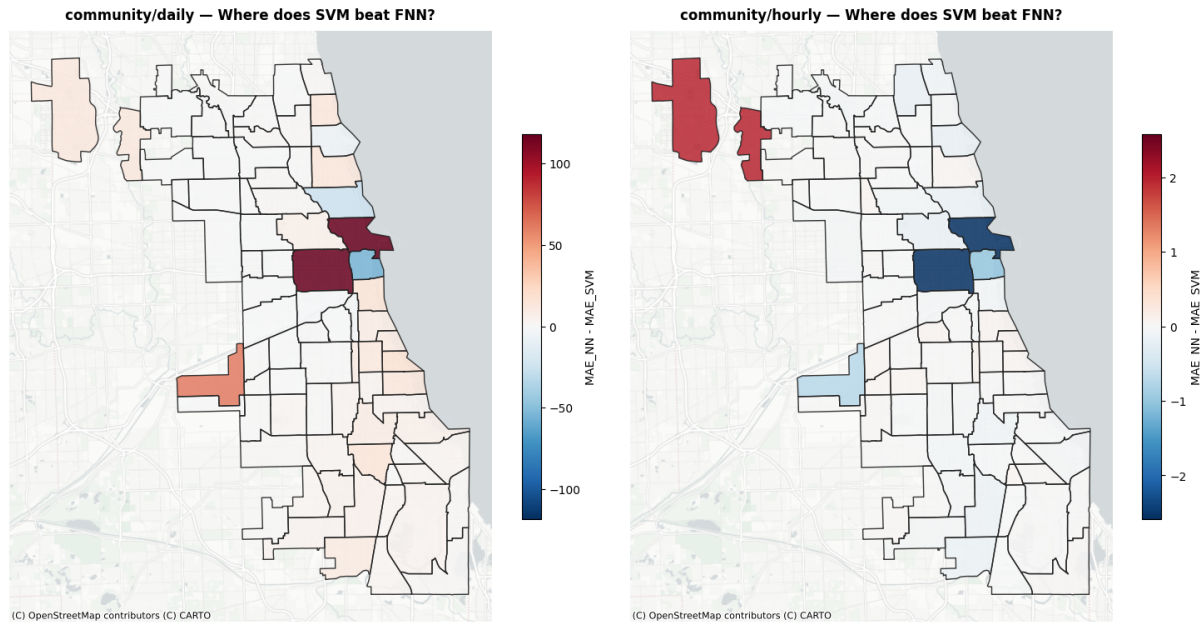


Figure 32: MAE_NN - MAE_SVM per community area per temporal resolution. Blue = FNN wins, red = SVM wins.

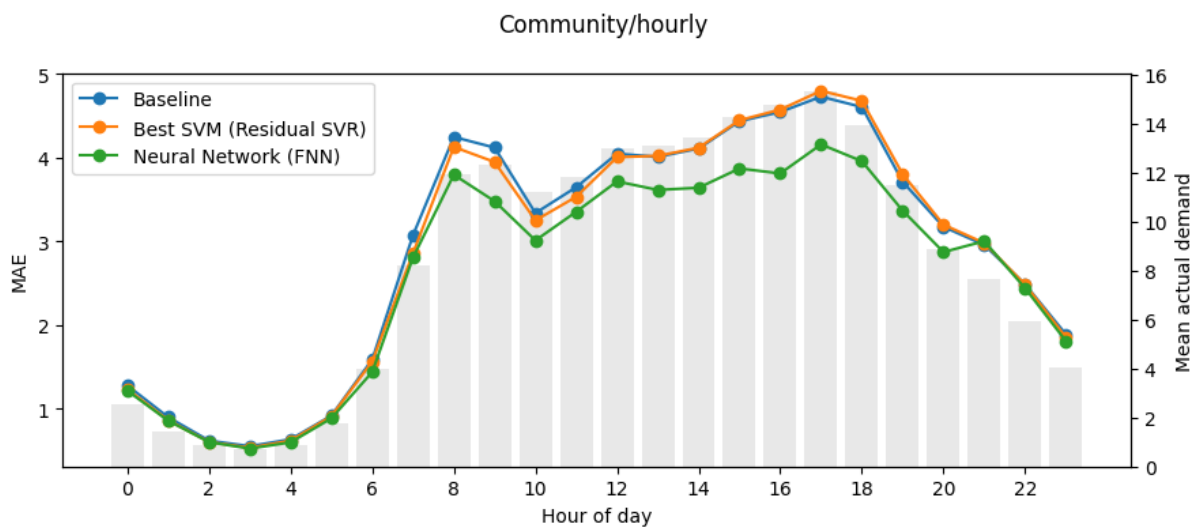


Figure 33: Test MAE by hour of day (Community/hourly).

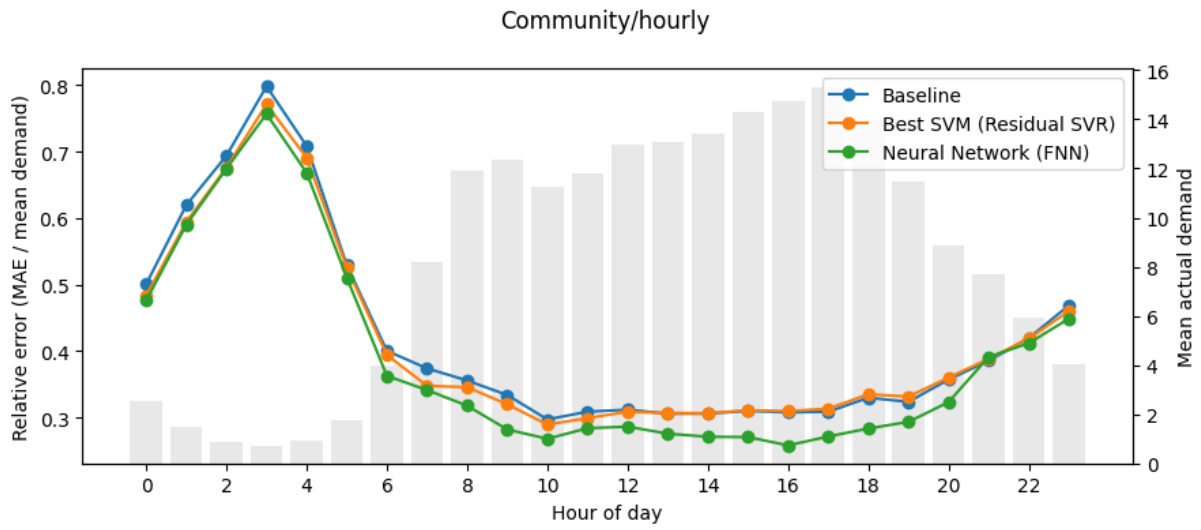
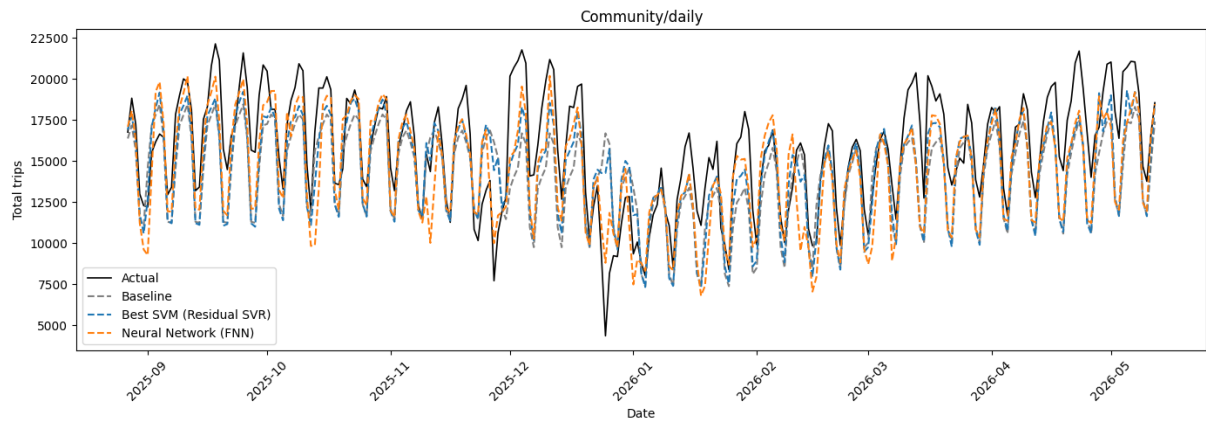
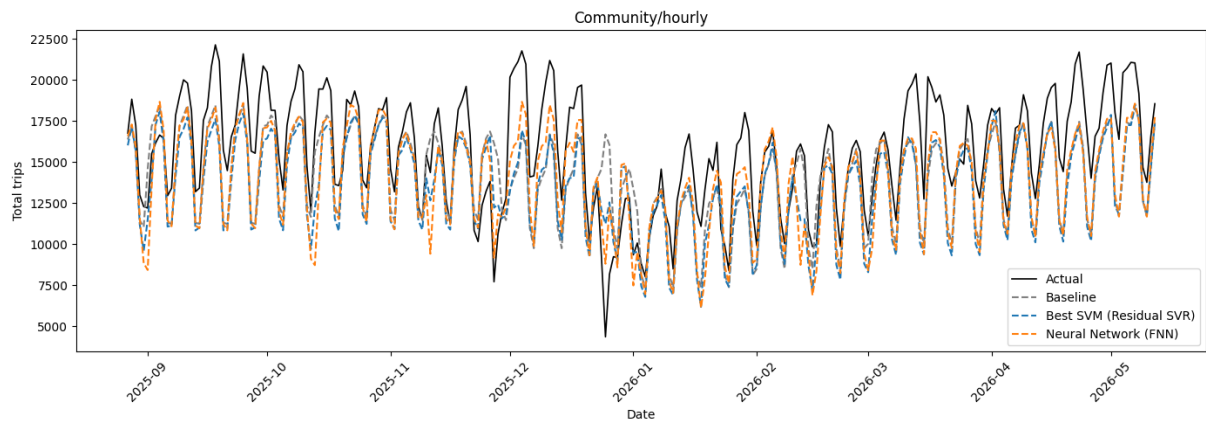


Figure 34: Test relative error (MAE / mean demand) by hour of day (Community/hourly).



(a)

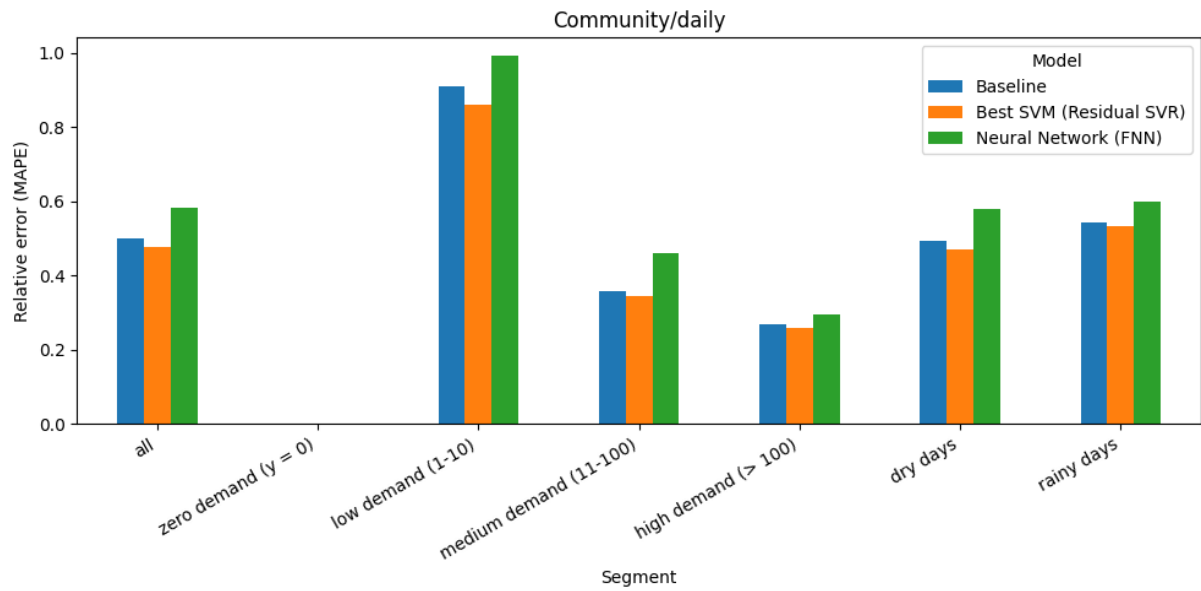


(b)

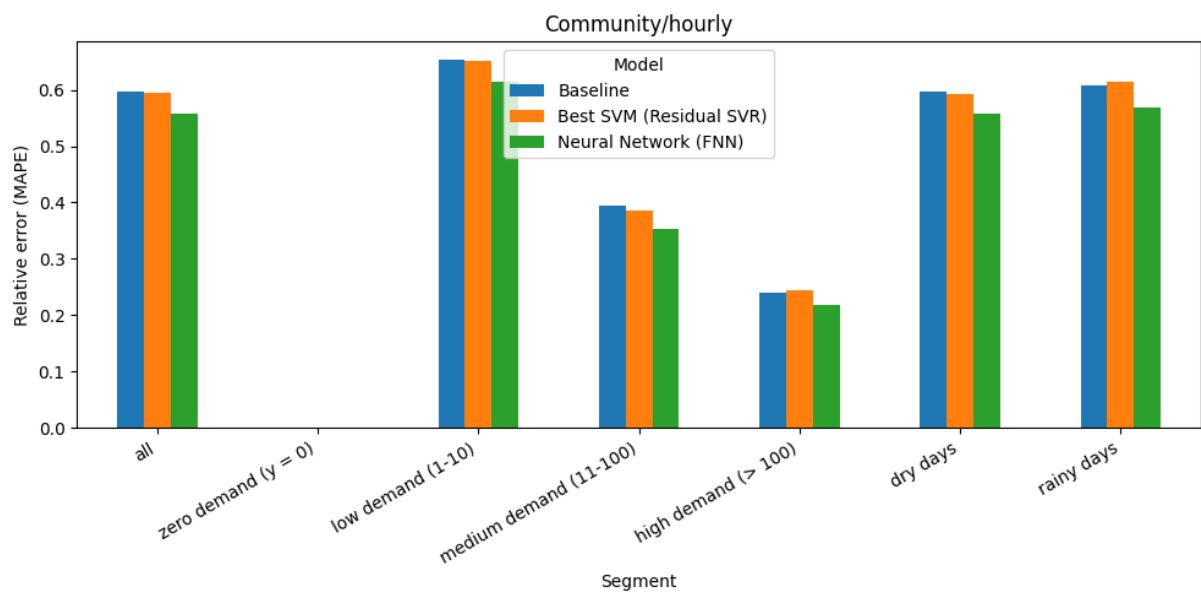
Figure 35: City-wide demand over the test period per temporal resolution.

Table 13: Test MAE, relative error, and skill score by demand band and weather per temporal resolution.

(a) community/daily		n	MAE	rel_error	skill_score
segment	Model				
all	Baseline	19943	49.575	0.499	0.000
	Best SVM (Residual SVR)	19943	46.337	0.476	0.065
	Neural Network (FNN)	19943	54.518	0.582	-0.100
zero demand ($y = 0$)	Baseline	250	2.873	NaN	0.000
	Best SVM (Residual SVR)	250	2.721	NaN	0.053
	Neural Network (FNN)	250	2.893	NaN	-0.007
low demand (1-10)	Baseline	5557	3.790	0.911	0.000
	Best SVM (Residual SVR)	5557	3.598	0.862	0.051
	Neural Network (FNN)	5557	4.391	0.992	-0.158
medium demand (11-100)	Baseline	10788	10.166	0.359	0.000
	Best SVM (Residual SVR)	10788	9.819	0.345	0.034
	Neural Network (FNN)	10788	13.234	0.460	-0.302
high demand (> 100)	Baseline	3348	256.040	0.267	0.000
	Best SVM (Residual SVR)	3348	238.203	0.260	0.070
	Neural Network (FNN)	3348	274.604	0.294	-0.073
dry days	Baseline	17633	49.678	0.493	0.000
	Best SVM (Residual SVR)	17633	46.481	0.469	0.064
	Neural Network (FNN)	17633	54.490	0.579	-0.097
rainy days	Baseline	2310	48.789	0.544	0.000
	Best SVM (Residual SVR)	2310	45.238	0.533	0.073
	Neural Network (FNN)	2310	54.735	0.601	-0.122
(b) community/hourly		n	MAE	rel_error	skill_score
segment	Model				
all	Baseline	478632	2.903	0.597	0.000
	Best SVM (Residual SVR)	478632	2.878	0.595	0.009
	Neural Network (FNN)	478632	2.620	0.559	0.097
zero demand ($y = 0$)	Baseline	230578	0.507	NaN	0.000
	Best SVM (Residual SVR)	230578	0.409	NaN	0.194
	Neural Network (FNN)	230578	0.428	NaN	0.156
low demand (1-10)	Baseline	202066	1.379	0.653	0.000
	Best SVM (Residual SVR)	202066	1.389	0.651	-0.007
	Neural Network (FNN)	202066	1.349	0.613	0.022
medium demand (11-100)	Baseline	33792	12.172	0.394	0.000
	Best SVM (Residual SVR)	33792	11.927	0.387	0.020
	Neural Network (FNN)	33792	10.801	0.353	0.113
high demand (> 100)	Baseline	12196	47.763	0.240	0.000
	Best SVM (Residual SVR)	12196	49.147	0.243	-0.029
	Neural Network (FNN)	12196	42.466	0.219	0.111
dry days	Baseline	421036	2.868	0.596	0.000
	Best SVM (Residual SVR)	421036	2.849	0.592	0.007
	Neural Network (FNN)	421036	2.590	0.557	0.097
rainy days	Baseline	57596	3.157	0.607	0.000
	Best SVM (Residual SVR)	57596	3.084	0.614	0.023
	Neural Network (FNN)	57596	2.837	0.569	0.101



(a)



(b)

Figure 36: Test relative error (MAE / mean demand) by segment per temporal resolution.

A.11 Reinforcement Learning (4.1)

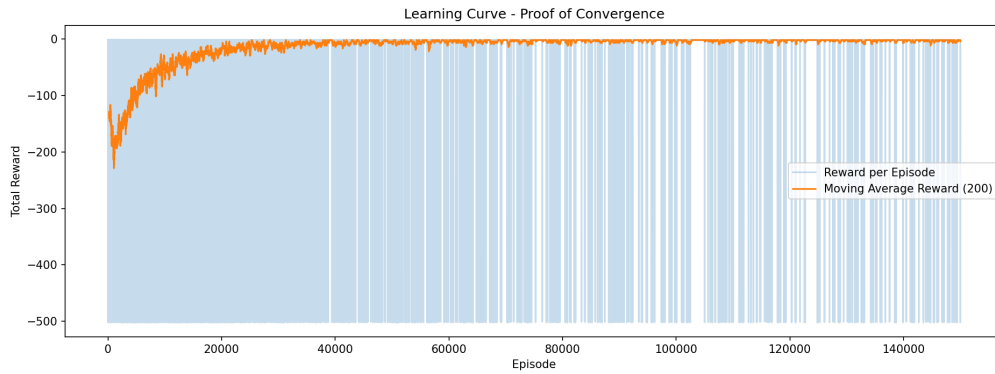


Figure 37: Figure 1a: Training score over time.

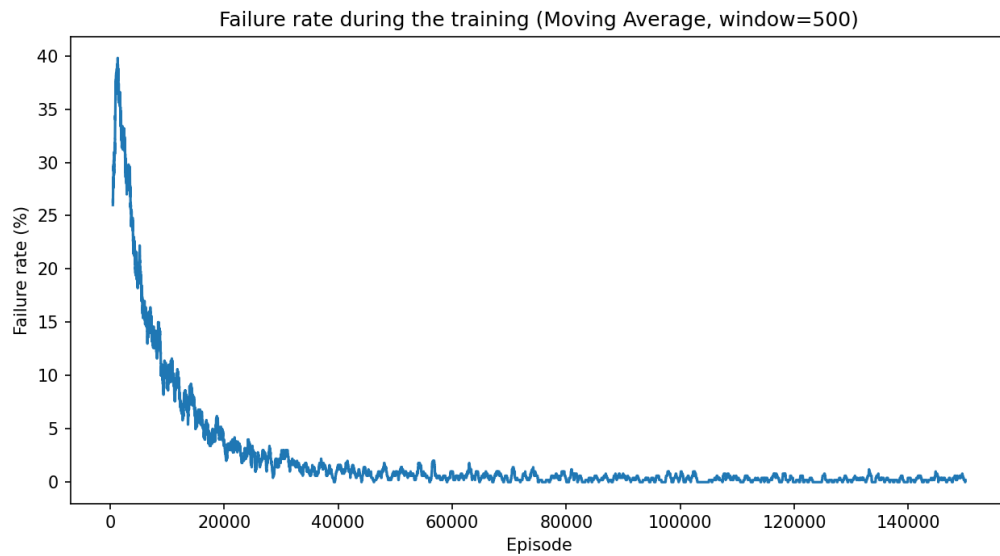


Figure 38: Figure 1b. Probability of running out of energy over time.

--- Calculating Naive Strategy Baseline (2000 days) ---

Naive Success Rate: 100.00%

Average Naive Charging Cost per day: \$1.92

Table 14: Table Naive vs. Tabular Q-Learning.

Strategy	Avg. Daily Cost	Probability of Running Out
0 Naive (always full power)	\$1.92	~0.14%
1 Tabular Q-Learning	\$1.50 $\pm$ \$0.02	0.11% $\pm$ 0.09%

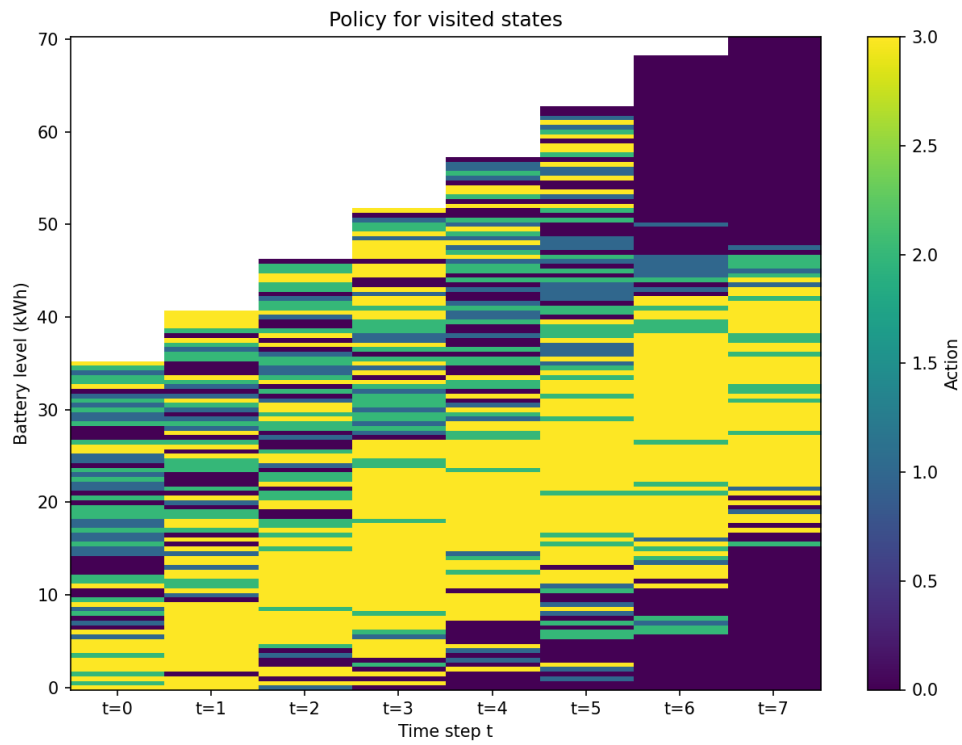


Figure 39: Figure 2. Learned charging policy. The color indicates the chosen charging speed for each battery-level/time combination.

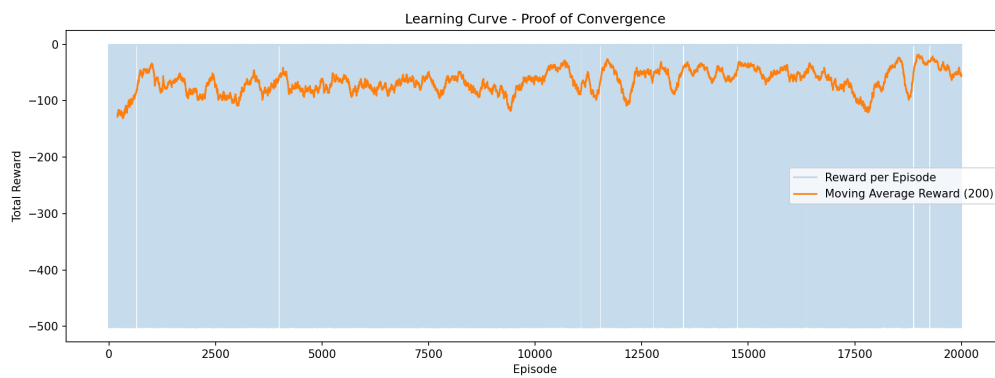


Figure 40: Figure 3a. DQN training score.

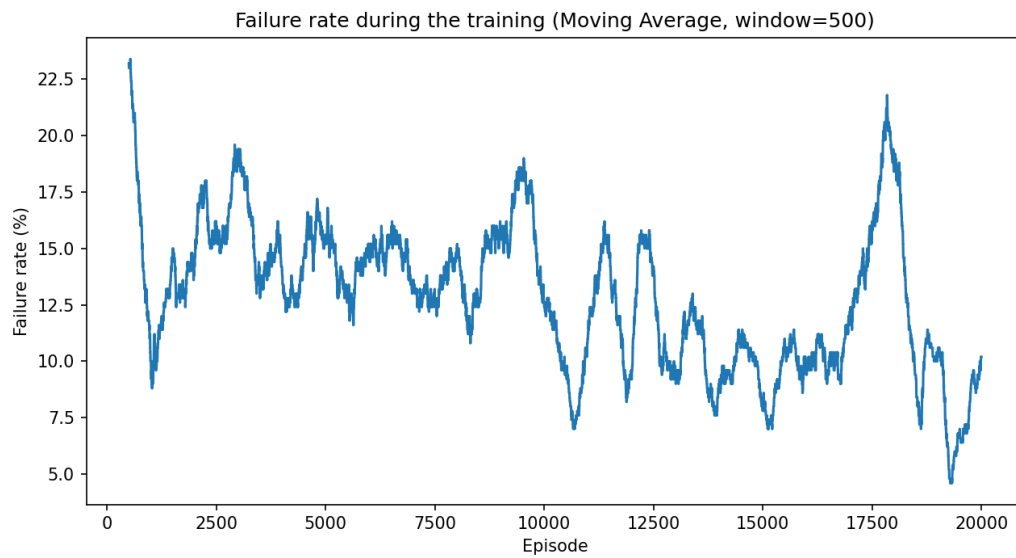


Figure 41: Figure 3b. DQN failure rate during training.

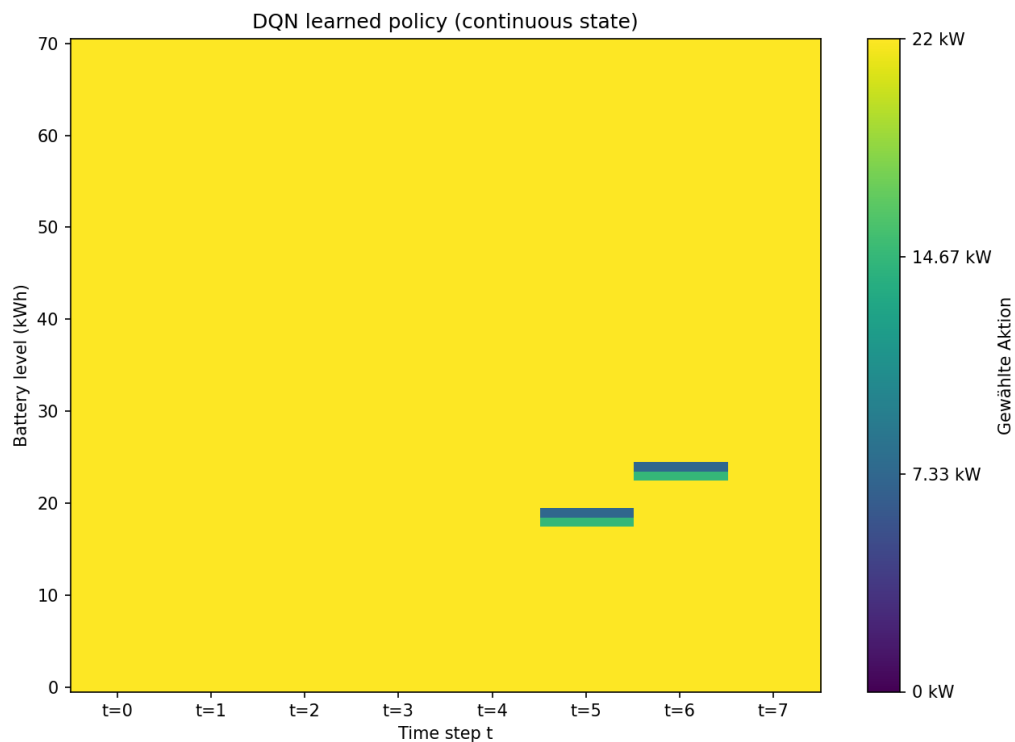


Figure 42: Figure 4. Best DQN policy found across the five training runs. The Q-table stores entries only for states that were actually visited during training ($\text{visit_counts} > 0$) and thus, combinations of time step and battery level that were never reached remain empty. The DQN network, by contrast, is a continuous function that produces an output for any given input. This includes states that never occurred during training (e.g. battery=60 at $t=0$). Thereby, both plots differ in terms of their appearance.

Table 15: Table DQN Summary.

Strategy	Avg. Daily Cost	Probability of Running Out
0 Deep Q-Network (DQN)	\$1.46 $\pm$ \$0.24	1.16% $\pm$ 0.95%

Table 16: Table DQN Per-Seed Breakdown.

Run (seed)	Avg. Daily Cost	Probability of Running Out	Best checkpoint found at episode
0 0	\$1.29	2.82%	16,000
1 1	\$1.92	0.00%	8,000
2 2	\$1.28	1.48%	18,000
3 3	\$1.53	0.70%	12,000
4 4	\$1.31	0.78%	12,000

References

- Loshchilov, I., & Hutter, F. (2019). Decoupled weight decay regularization. *7th International Conference on Learning Representations, ICLR 2019*.
- Yan, J., Xiang, L., Wu, C., & Wu, H. (2020). City-scale taxi demand prediction using multisource urban geospatial data. *The International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences*, 43, 213–220.